- Welcome!!

Fair, Accountable, Transparent Machine Learning (FAccT ML)
ZG517

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Session 8
Time – 10:40 AM to 12:40 PM

These slides are prepared by the instructor, with grateful acknowledgement of many others who made their course materials freely available online.

Midsem Tips

- 30 marks : 3/4 questions
  - 25-35% theory questions
    - → Rest numerical
      - → Calculator!!
- Pattern similar to past papers!!
  Numerical
- Expect basic knowledge about
  ML, MFML/ISM, & some basics
  of DNN, gradient descent

Midsem Tips

Summary

- Fair Machine Learning
  - Prevents ML models from biasing toward specific groups when allocating favorable outcomes
- Group Treatments of Fairness Metrics
  - Demographic Parity
  - Equalized Odds/Opportunity
- Individual Treatments of Fairness
  - Fairness Through Awareness Individual Fairness
  - Individual Fairness
  - Counterfactual Fairness
- Fair ML Techniques must read paper Kamiran
  - Pre-processing Methods: Resampling, Reweighting, Optimized-preprocessing elaborate
  - In-processing Methods: Regularization, Adversarial Learning
  - Post-processing Methods: Learning to Defer

Summary

Digital

- Fair NLP Methods
  - Debiasing Word Embeddings
  - Data Augmentation
    - Gender Swapping
  - Fair Representation for Pre-trained Encoders

Completes our discussion on fairness

'CV'

- Fair Visual Representations
  - Counterfactual Face Attribution
  - Gender Equalized Image Captioning
  - Adversarial Removal of Gender Features

Contd 4-5 on interpretability/explainability
2-3 on Security/privacy

Sensitive (Protected) Features

- Sensitive Features
  - Identify a group
  - e.g., gender, ethnics
- Discrimination Occurs
  - When Sensitive Features Are Used Improperly
  - May lead to ML Discrimination

disease mut → Y

two

gene(s) → of

trade off between

ML accuracy / Fairness

"could be"

Fairness Through Unawareness

- A ML Algorithm Achieves Fair Through Unawareness If
  - None of the sensitive features are directly used in the model

Protected

~~Race and Ethnicity~~	Skills	Years of Exp	Hired?
~~Hispanic~~	Javascript	1	no
~~Hispanic~~	C++	5	yes
~~White~~	Java	2	yes
~~White~~	C++	3	yes

Training

Fair ML Model

Issues With Fairness Through Unawareness

Sensitive Features May Still Be Used

- Inferred from indirect evidence

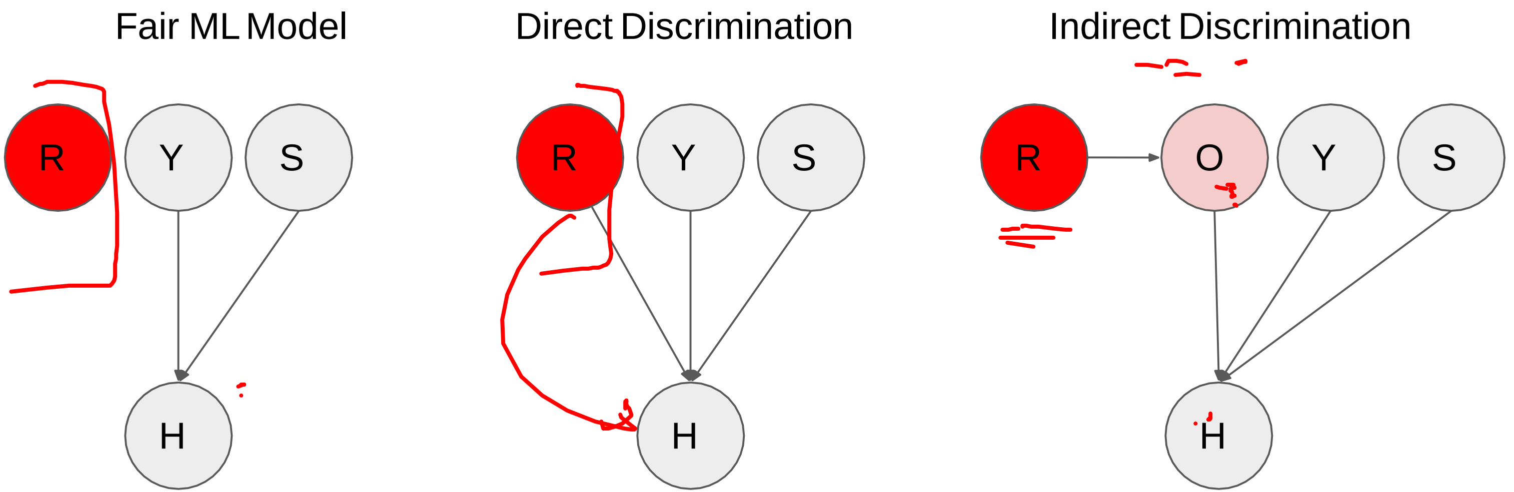
Inferred

Protected

~~Race and Ethnicity~~	Skills	Years of Exp	Often Goes to Mexican Markets	Hiring Decision
~~Hispanic~~	Javascript	1	yes	no
~~Hispanic~~	C++	5	yes	yes
~~White~~	Java	2	no	yes
~~White~~	C++	3	no	yes

Training → Discriminatory ML Model

Types of Discriminations



R - Race
Y - Years of Exp

S = Skills
O = Often Goes to Mexico Market

more natural/expected

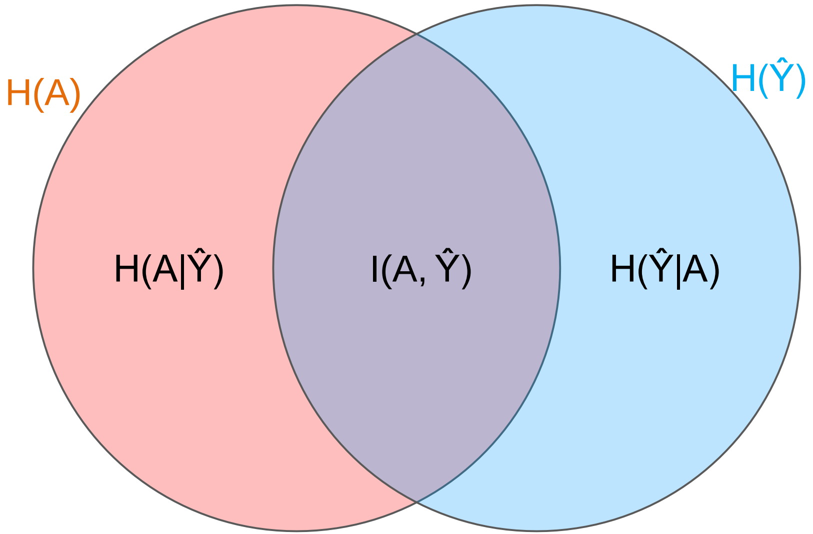
Conditions for Direct Discrimination

- A - set of protected features
- X - set of features other than protected features

- A predictor Ŷ is direct discrimination if
  - $P(\hat{Y} \mid X, A) \neq P(\hat{Y} \mid X)$
  - i.e., $\hat{Y} \not\perp A \mid X$

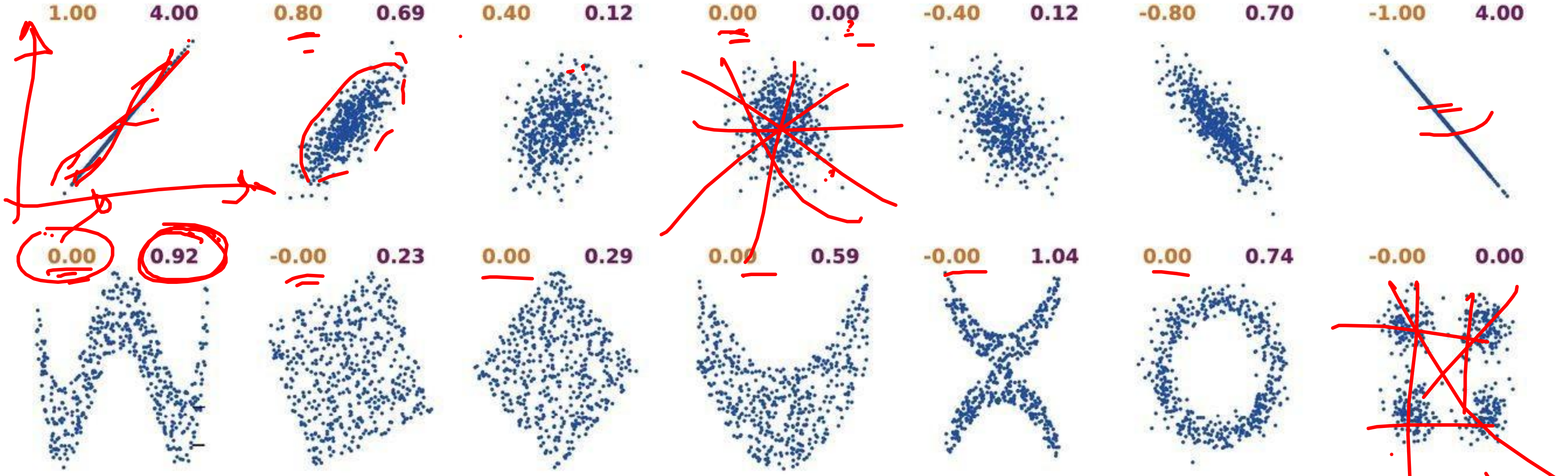
Conditions for Indirect Discrimination

- Mutual Information
    - A measure of the mutual dependence between A and $\hat{Y}$
    - $I(A, \hat{Y}) = H(A) - H(A \mid \hat{Y}) = H(\hat{Y}) - H(\hat{Y} \mid A)$
    - $I(A, \hat{Y}) = 0$ if $P(\hat{Y} \mid A) = P(\hat{Y})$, or $A \perp\!\!\!\perp \hat{Y}$



Correlation Coefficient and Mutual Information

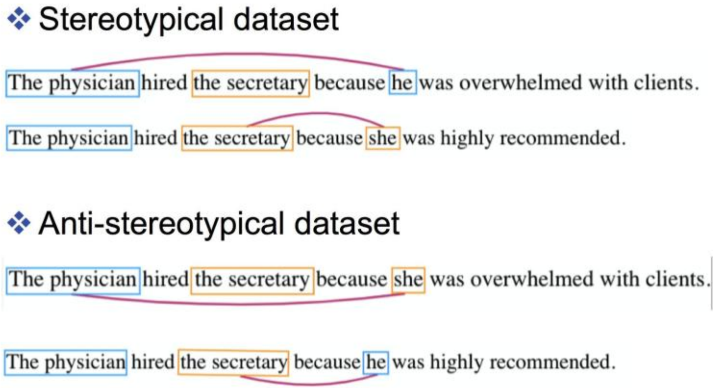
Correlation Coefficient (left) and Mutual Information (right)



Ince et al, 2016

Limitations

- Processing Sensitive Features
  - Fairness through unawareness requires sensitive features to be masked out
  - Not easy to do in real life
  - Referred to as individual fairness criteria



Common Fairness Criteria

- Demographic Parity
- Equality of Odds
- Equality of Opportunity

Demographic Parity Applied to a Group

Demographic Parity Is Applied to a Group of Samples

– Does not require features to be masked out

A Predictor Ŷ Satisfies Demographic Parity If

– The probabilities of positive predictions are the same regardless of whether the group is protected
– Protected groups are identified as A = 1

$$P(\hat{Y} = 1 | A = 1) = P(\hat{Y} = 1 | A = 0)$$

expected output
+ve

$P(\hat{Y}=1|M)$
30/100

M
100 → A +ve 30

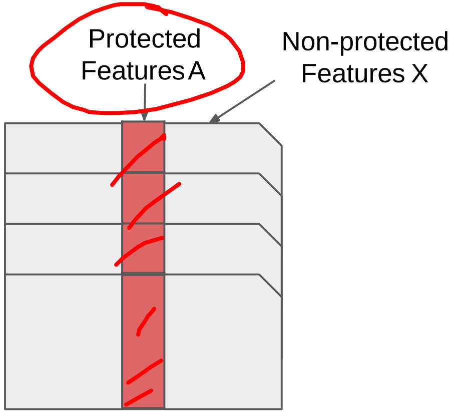
F
50 → 15

$P(\hat{Y}=1|F) = \frac{15}{50} = 0.3 =$

Comparisons

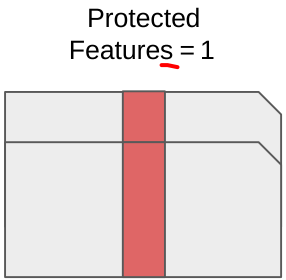
Individual Treatment

Group Treatment



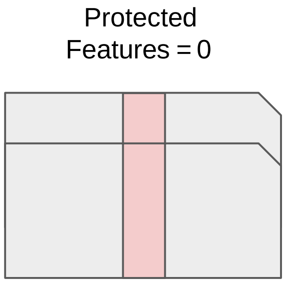
Fairness Through Unawareness

$P(\hat{Y} \mid X)$



Demographic Parity

$P(\hat{Y}=1 \mid A=1)$



Demographic Parity

$P(\hat{Y}=1 \mid A=0)$

Equality of Odds

Equal Probabilities for Both Qualified/Unqualified People Across Protected Groups

P(Ŷ=1|Y=0)

A=1

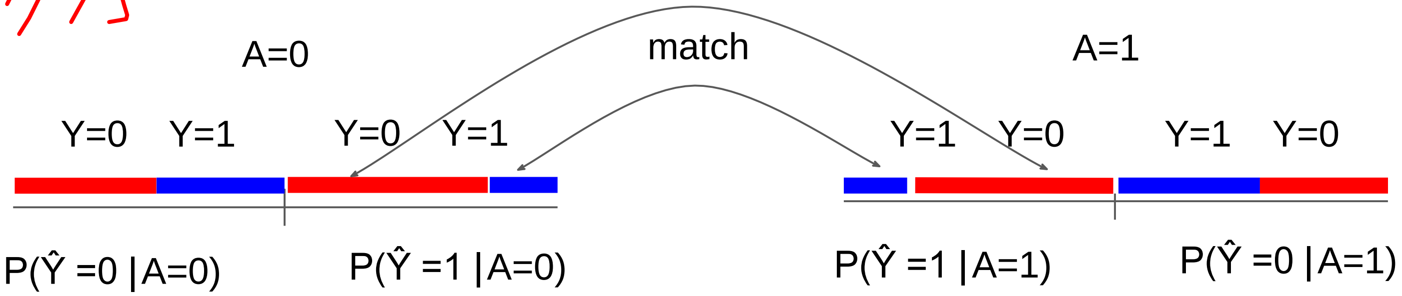
A=0

P(Ŷ=1|Y=0)

$$P(\hat{Y} = 1|A = 0, Y) = P(\hat{Y} = 1|A = 1, Y)$$

probability

ground truths



Hardt et al, 2016

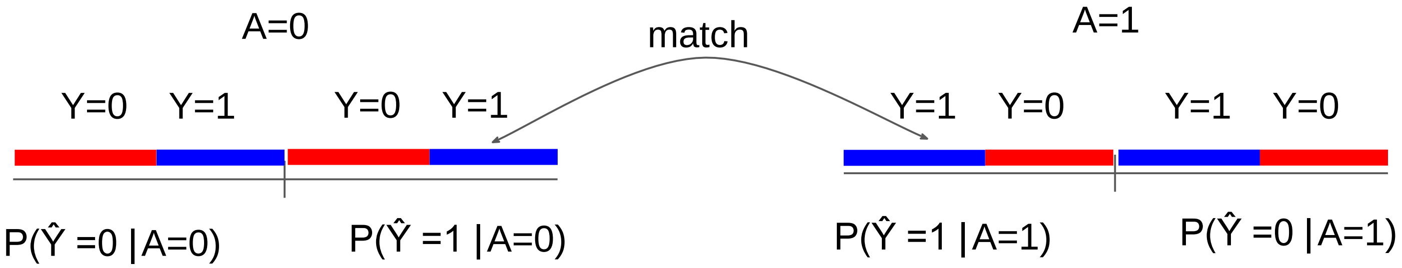
Equality of Opportunity

Equal Probabilities for Qualified People Across Protected Groups

Y=0

$$P(\hat{Y}=1|\boxed{A=0},Y=1)=P(\hat{Y}=1|\boxed{A=1},Y=1)$$

Y≠1



Hardt et al, 2016

Practice Question

Find out the Fairness Criteria that Ŷ1, and Ŷ2 Satisfy

– A = {race}, Y = {Hiring Decision}

Race and Ethnicity	Skill	Years of Exp	Goes to Mexican Markets?	Hiring Decision Y	Predictor $\hat{Y}_1$ predicted	Predictor $\hat{Y}_2$
Hispanic	Javascript	1	yes	no	0	1
Hispanic	C++	5	yes	yes	1	1
Hispanic	Python	1	no	yes	1	0
White	Java	2	no	yes	0	0
White	C++	3	no	yes	1	1
White	C++	0	no	no	1	0

Demographic Parity for Predictor Ŷ1

$P(\hat{Y}1 = 1 \mid R = H) = 2/3$

$P(\hat{Y}1 = 1 \mid R = W) =$ 2/3

Race and Ethnicity	Skill	Years of Exp	Goes to Mexican Markets?	Hiring Decision Y	Predictor $\hat{Y}_1$	Predictor $\hat{Y}_2$
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Demographic Parity for Predictor $\hat{Y}1$

- $P(\hat{Y}1 = 1 \mid R = H) = 2/3$
- $P(\hat{Y}1 = 1 \mid R = W) = 2/3$

✔ Demographics Parity

$$P(\hat{Y} = 1|A = 1) = P(\hat{Y} = 1|A = 0)$$

Race and Ethnicity	Skill	Years of Exp	Goes to Mexican Markets?	Hiring Decision Y	Predictor $\hat{Y}_1$	Predictor $\hat{Y}_2$
Hispanic	Javascript	1	yes	no	0	1
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Equality of Opportunity/Odds for Ŷ1

- $P(\hat{Y}1 = 1 \mid R = H, Y = yes) = 1$ = 2/2 ≠ E. Oppty not achieved ⇓ E. odds not achieved
- $P(\hat{Y}1 = 1 \mid R = W, Y = yes) =$ 1/2
- $P(\hat{Y}1 = 1 \mid R = H, Y = no) =$ 0
- $P(\hat{Y}1 = 1 \mid R = W, Y = no) =$ 1/2 ≠

Race and Ethnicity	Skill	Years of Exp	Goes to Mexican Markets?	Hiring Decision Y	Predictor $\hat{Y}_1$	Predictor $\hat{Y}_2$
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Equality of Opportunity/Odds for $\hat{Y}1$

- $P(\hat{Y}1 = 1 \mid R = H, Y = yes) = 1$
- $P(\hat{Y}1 = 1 \mid R = W, Y = yes) = 0.5$
- $P(\hat{Y}1 = 1 \mid R = H, Y = no) =$
- $P(\hat{Y}1 = 1 \mid R = W, Y = no) =$

Race and Ethnicity	Skill	Years of Exp	Goes to Mexican Markets?	Hiring Decision Y	Predictor $\hat{Y}_1$	Predictor $\hat{Y}_2$
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- $P(\hat{Y}1 = 1 \mid R = W, Y = no) = 1$

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Equality of Opportunity/Odds for Ŷ1

- $P(\hat{Y}1 = 1 \mid R = H, Y = yes) = 1$
- $P(\hat{Y}1 = 1 \mid R = W, Y = yes) = 0.5$
- $P(\hat{Y}1 = 1 \mid R = H, Y = no) = 0$
- $P(\hat{Y}1 = 1 \mid R = W, Y = no) = 1$

Equality of Opportunity

$$P(\hat{Y} = 1 | A = 0, Y = 1) = P(\hat{Y} = 1 | A = 1, Y = 1)$$

Equality of Odds

$$P(\hat{Y} = 1 | A = 0, Y) = P(\hat{Y} = 1 | A = 1, Y)$$

$1 - 1 < \epsilon$ $\boxed{0.1 - 0.2}$

Race and Ethnicity	Skill	Years of Exp	Goes to Mexican Markets?	Hiring Decision Y	Predictor $\hat{Y}_1$	Predictor $\hat{Y}_2$
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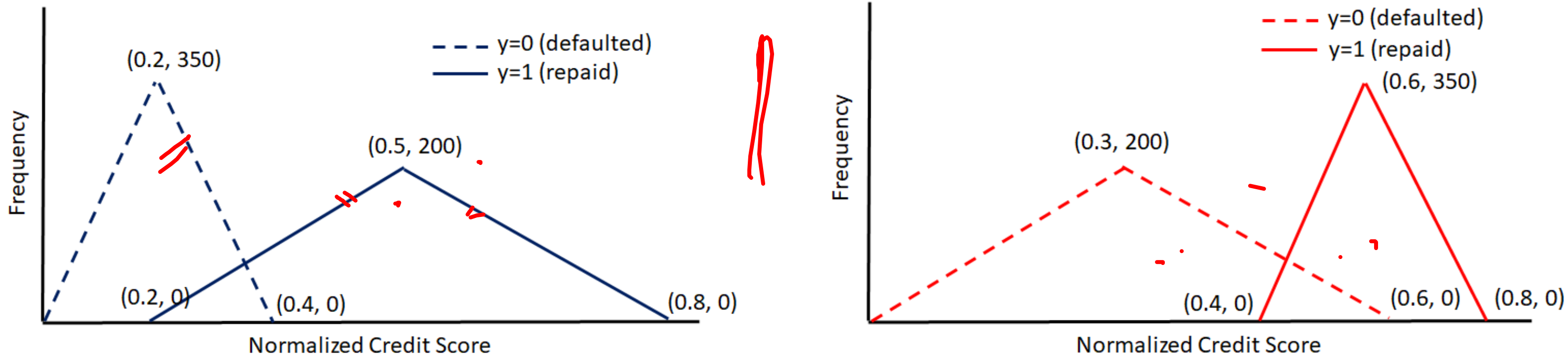
Summary of Fairness Criteria

Fairness Criteria	Criteria	Group	Individual
Unawareness	Excludes A in Predictions		✓
Demographic Parity	$P(\hat{Y}=1\|A=0) = P(\hat{Y}=1\|A=1)$	✓	
Equalized Odds	$P(\hat{Y}=1\|A=0,Y) = P(\hat{Y}=1\|A=1,Y)$	✓	
Equalized Opportunity	$P(\hat{Y}=1\|A=0,Y=1) = P(\hat{Y}=1\|A=1,Y=1)$	✓	

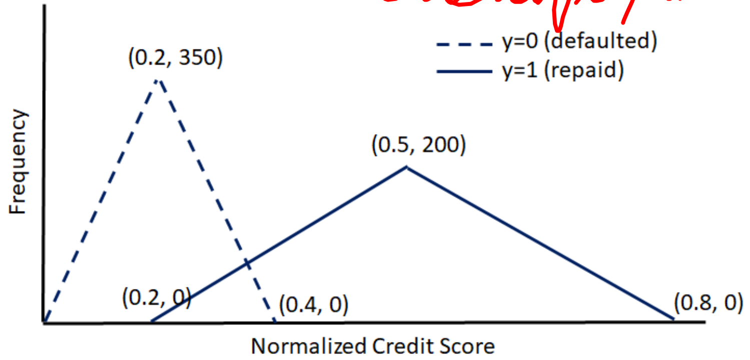
ML EC-3M Problem

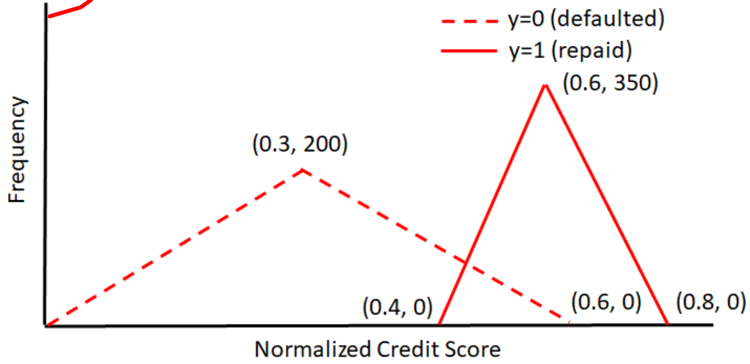
Consider a financial institution that uses normalized credit score for approving or rejecting housing loan. Note there are two population of applicants 'blue' (deprived group with protected attribute A=0) and 'red' (favored group with A=1). Loan is approved, i.e., y'=1 if normalized credit score is greater than some threshold t, else rejected. y=1 indicates approved loans that are repaid based on historical data, and y=0 indicates approved loan was defaulted. The financial institution wants to approve loans that is likely to be repaid. Distributions are described in following figure (not drawn to scale) for both populations.



A. Calculate probability P(y′=1) for both ‘blue’ and ‘red’ populations for threshold t=0.4. Is fairness achieved w.r.t. protected attribute A?

demographic parity





For ‘blue’ population,

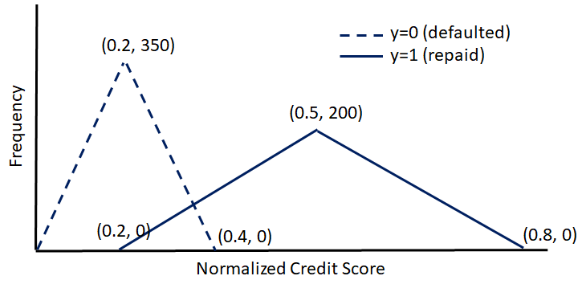
P(y′=1)=[0.5*200*0.6-0.5*0.2*133.3]/[0.2*350+0.3*200]= 0.359

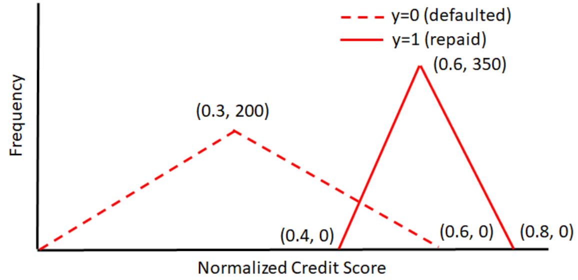
For ‘red’ population,

P(y′=1)=[0.2*350+0.5*0.2*133.3]/[0.3*200+0.2*350]= 0.641 ➔ Fairness not achieved.

ML EC-3M Problem

B. If threshold t=0.45 is chosen for the blue population, calculate the probability P(y'=1) for the blue population. What threshold value for the red population will ensure demographic parity?





For 'blue' population, for t=0.45

P(Y'=1)=[0.5*200*0.6-0.5*0.25*166.67]/[0.2*350+0.3*200]= 0.30

For 'red population, for any 0.4<t<0.6

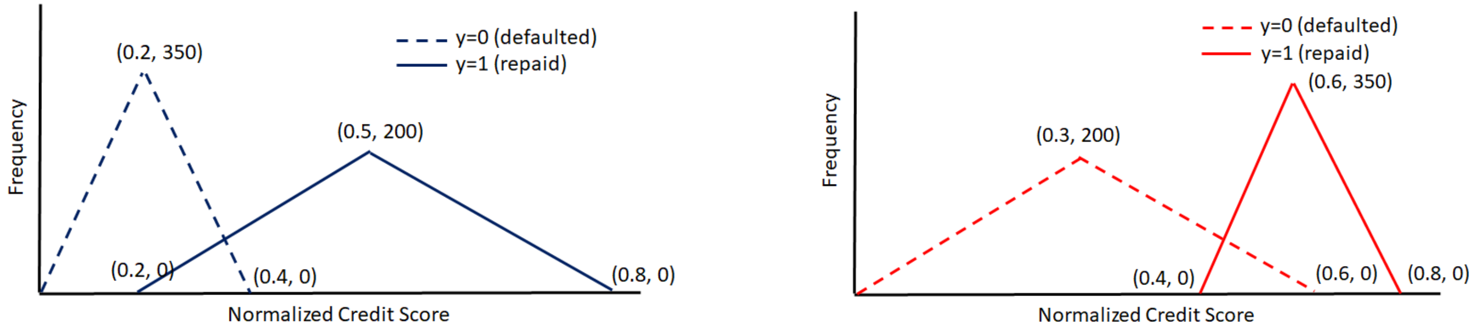
P(Y'=1)=[0.2*350+0.5*(0.6-t)*200/0.3*(0.6-t)-0.5*(t-0.4)*350/0.2]/[0.3*200+0.2* 350]

Therefore, for demographic parity,

$70+0.5*(0.6-t)^2*333.33-0.5*(t-0.4)*350/0.2=39$ ➔ t=0.44463

ML EC-3M Problem

C. If threshold t=0.5 is chosen for the blue population, calculate the probability P(y'=1|y=1). What threshold value for the red population will ensure equal opportunity?



For the blue population, from the frequency distribution

- ✓ for t=0.5 P(y'=1|y=1)=0.5

For the red population, from the distribution figure, for t=0.6 P(y'=1|y=1)=0.5

Fair ML Methods

~~List~~ Distribution of input data

- **Pre-processing Methods** →
  - **Transform data before ML models learn**
  - **e.g., Reweighting, Resampling**
  - **Additional Optimization**

- In-processing Methods
  - Constrain ML models while they learn, e.g.,
    - Adversarial Learning
    - Prejudice Removing Regularizer,

- Post-processing Methods
  - Make predictions from a black-box ML model fair in the post-processing stage
  - e.g., Learning to Defer

Basic Data Manipulation Techniques

• Massaging → changing range output labels according to some

- Reweighing
- Universal Sampling
- Preferential Sampling

Scaling

Fav over sampling

ranking fn

deletes

Deprived

Favorable

over

Biased Data

Observed: M = 10, F = 4

Expected: M = 7, F = 7

Expected Distribution of Fair Data

- Expected Data Distribution

$$P(Y) = P(Y|A=1) = P(Y|A=0)$$

which leads to $Y \perp\!\!\!\perp A$

- Recall Demographic Parity

$$P(\hat{Y}=1|A=1) = P(\hat{Y}=1|A=0)$$

[Kamiran et al, 2012](#)

Expected Distribution of Fair Data

- The Expected Joint Distribution Under $Y \perp\!\!\!\perp A$

outcome is ⊥ sensitivity

$$P_{exp}(Y = y, A = a) = P(Y = y) \cdot P(A = a)$$
$$= \frac{|\{x \in \mathcal{D} | x_Y = y\}|}{|\mathcal{D}|} \cdot \frac{|\{x \in \mathcal{D} | x_A = a\}|}{|\mathcal{D}|}$$

- Our Observed Joint Distribution

actual dataset

$$P_{obs}(Y = y, A = a) = \frac{|\{x \in \mathcal{D} | x_Y = y, x_A = a\}|}{|\mathcal{D}|}$$

Transform Data to Expected Distribution

[Kamiran et al, 2012]

Reweighting

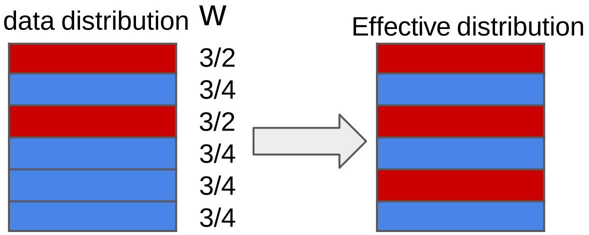
$L = \sum_{i=1}^{n} (\hat{y} - y_{act}) W(i)$

- Sample Weight for x
  - Goal: adjust our data to a distribution that leads to $Y \perp\!\!\!\perp A$, or Demographic Parity
  - W(x) = 1, we have achieved $Y \perp\!\!\!\perp A$ and Demographic Parity
  - W(x) > 1, increase the weight of sample x in training
  - W(x) < 1, decrease the weight of sample x in training

$$W(x) = \frac{P_{exp}(Y = x_y, A = x_a)}{P_{obs}(Y = x_y, A = x_a)}$$

- Reweighting Loss Function

$$\mathcal{L} = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} W(x) \cdot \mathcal{L}(\hat{Y}, x_Y)$$



Practice Question

- Calculate $W(x_3)$, A= {Sex}, Y = {Class}

Sex	Highest degree	Job type	Class
M	H. school	Board	+
M	Univ.	Board	+
M	H. school	Board	+
M	H. school	Healthcare	+
M	Univ.	Healthcare	–
F	Univ.	Education	–
F	H. school	Education	–
F	None	Healthcare	+
F	Univ.	Education	–
F	H. school	Board	+

Kamiran et al, 2012

Practice Question

p(A) = 0.5

p(+) = 6/10 = 0.6

- $W(x_3)$
  - $A_3 = M$
  - $Y_3 = +$
- Expected Distribution
  - $P(A = M) = 0.5$
  - $P(Y = +) = 0.6$
  - $P_{exp}(A = M, Y = +) = \boxed{0.3}$ = 0.5 * 0.6
- Observed Distribution
  - $P_{obs}(A = M, Y = +) = 0.4$

0.4 → 0.3

- Sample Weight
  - $W(x_3) = \boxed{0.3/0.4} = \boxed{0.75}$

you define

Sex, Highest deg, Job type, class

A = {Sex}, Y = {Class}

Sex	Highest degree	Job type	Class
M	H. school	Board	+
M	Univ.	Board	+
M	H. school	Board	+
M	H. school	Healthcare	+
M	Univ.	Healthcare	–
F	Univ.	Education	–
F	H. school	Education	–
F	None	Healthcare	+
F	Univ.	Education	–
F	H. school	Board	+

predicted

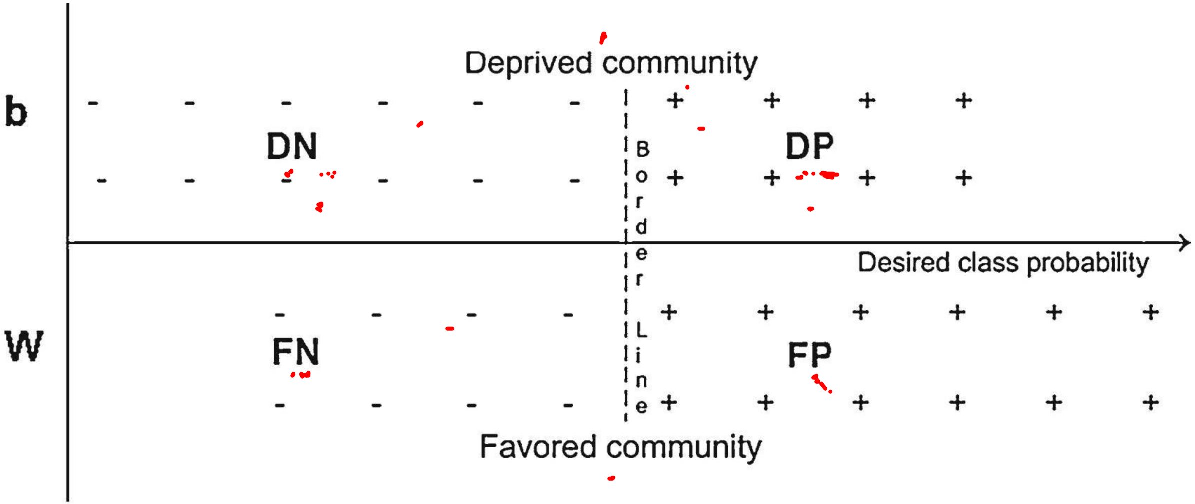
w_3

$\frac{1}{1+e^{-(W_0 - W_1 Sex - W_2 deg - W_3 type - 1)}}$

[Kamiran et al, 2012](#)

Resampling

- Resample the Dataset Based on the Expected Joint Probability



[Kamiran et al, 2012]

Expected Number of Samples

- Expected Number of Samples for the Category (y, a)

$$N_{exp}(y,a) = P_{exp}(y,a) \cdot |\mathcal{D}|$$

- Also Note

$$\sum_{y,a} N_{exp} = \sum_{y,a} P_{exp}(y,a) \cdot |\mathcal{D}| = |\mathcal{D}|$$

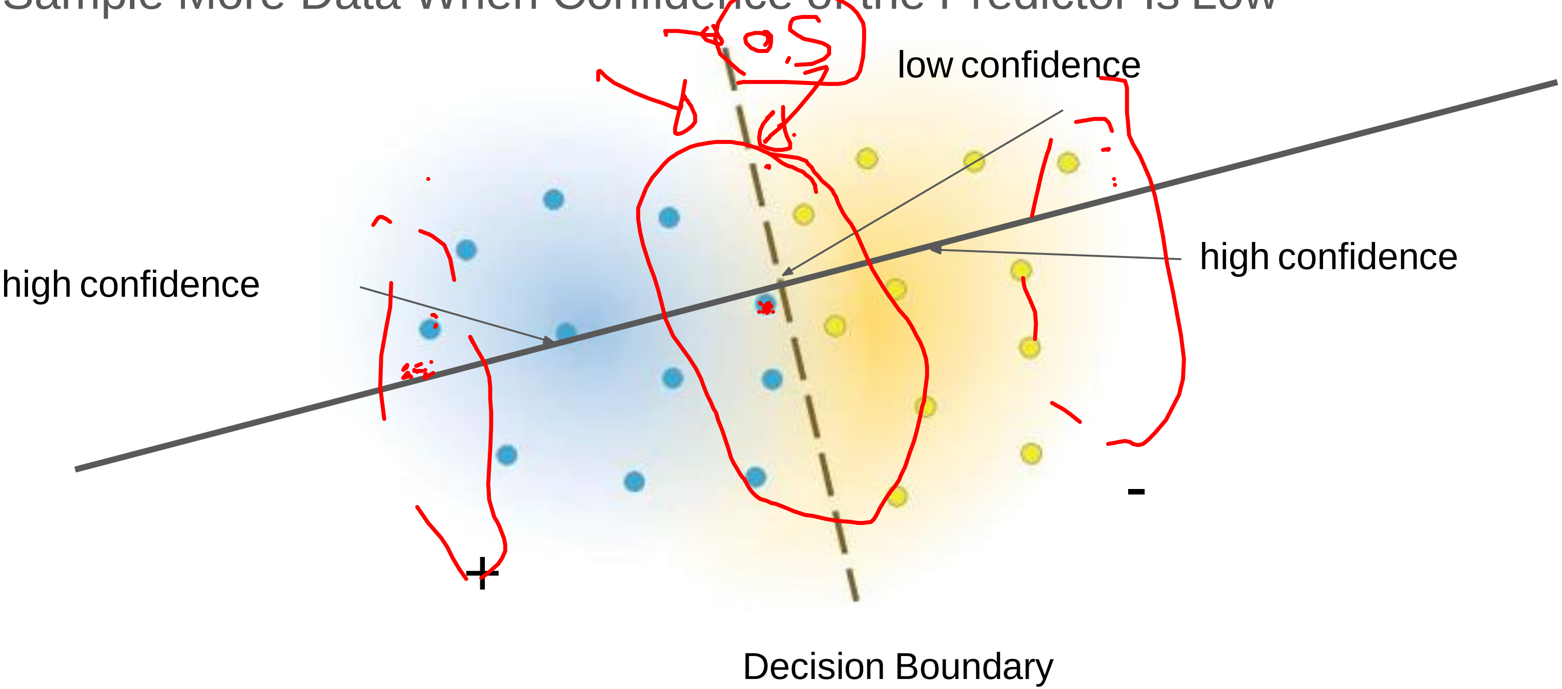
Universal Resampling (US)

- Resampling Based on the Expected Probabilities to Meet Demographic Parity
  - DP (Deprived community with Positive class labels)
    - draw $N_{exp}(D, P)$ samples uniformly from DP
  - DN (Deprived community with Negative class labels)
    - draw $N_{exp}(D, N)$ samples uniformly from DN
  - FP (Favored community with Positive class labels)
    - draw $N_{exp}(F, P)$ samples uniformly from FP
  - FN (Favored community with Negative class labels)
    - draw $N_{exp}(F, N)$ samples uniformly from FN

Kamiran et al, 2012

Preferential Sampling (PS)

- Sample More Data When Confidence of the Predictor Is Low



[Kamiran et al, 2012]

Bias Measures

- Measure prediction biases by comparing the favorable outcomes given to group 1 with that to group 0

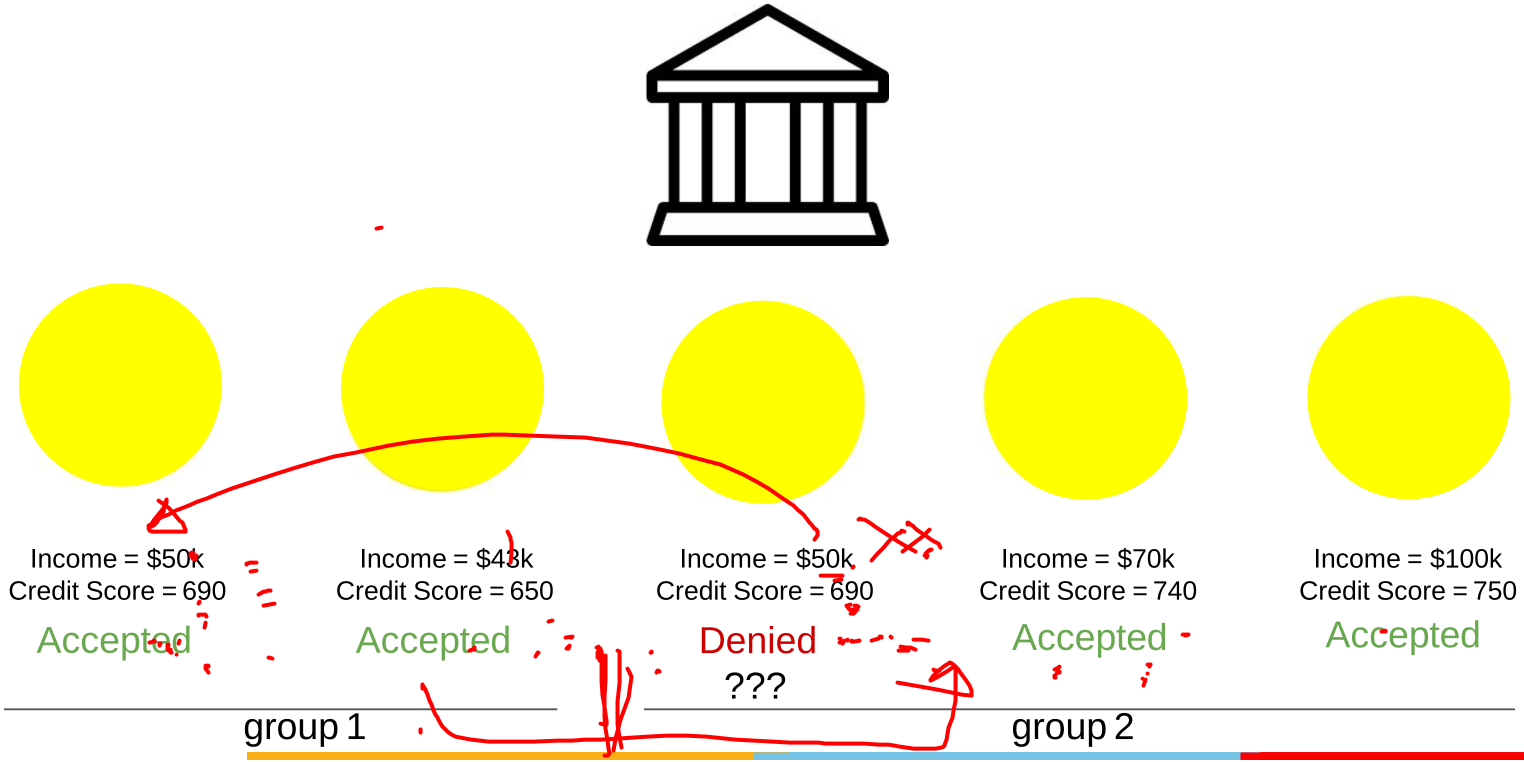
$$Bias(\hat{Y}) = P(\hat{Y} = 1|A = 1) - P(\hat{Y} = 1|A = 0)$$

Demographic Parity

$$P(\hat{Y} = 1|A = 1) = P(\hat{Y} = 1|A = 0)$$

[Kamiran et al, 2012](#)

Individual Fairness

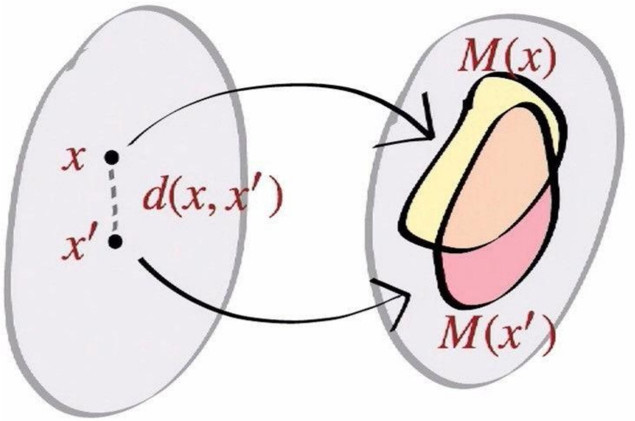


Individual Fairness

- A predictor M achieves individual fairness under a distance metric d iff
  - Similar Samples are treated similarly, in other words

w/o sensitive attributes

$$M(x_i) \approx M(x_j) | d(x_i, x_j) \approx 0$$



Individual Fairness

Individual

$$M(x_i) \approx M(x_j) | d(x_i, x_j) \approx 0$$

Income = $19k
Credit Score = 690

Income = $23k
Credit Score = 720

Income = $60k
Credit Score = 800

Group 1

Income = $20k
Credit Score = 680

Income = $27k
Credit Score = 700

Income = $65k
Credit Score = 810

Group 2

Fairness Criteria

Individual Treatment	Group Treatment
Fairness Through Unawareness Excludes Sensitive Information A from the predictor	Demographic Parity $P(\hat{Y}=1\|A=1)=P(\hat{Y}=1\|A=0)$
Individual Fairness $M(x_i) \approx M(x_j)\|d(x_i,x_j) \approx 0$	Equal Opportunity/Odds $P(\hat{Y}=1\|A=0,Y=1)=P(\hat{Y}=1\|A=1,Y=1)$ $P(\hat{Y}=1\|A=0,Y)=P(\hat{Y}=1\|A=1,Y)$

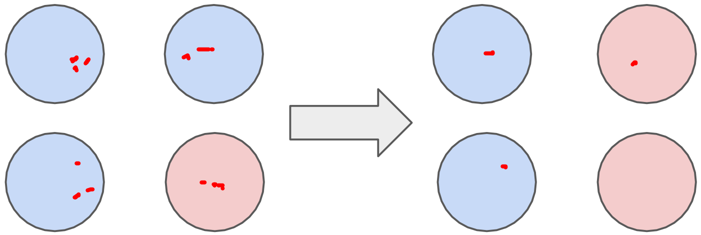
Optimized Pre-Processing for Fairness

- Can We Automate the Resampling Process?
  - Turn the the manual process into an optimization based approach
  - Include more criteria than Demographic Fairness
  - Allow transformations of data
- Optimized Pre-Processing
  - Given sensitive feature D, learn a probabilistic mapping $p_{\hat{X},\hat{Y}|X,Y,D}$ that transfers
  - Satisfies three constraints

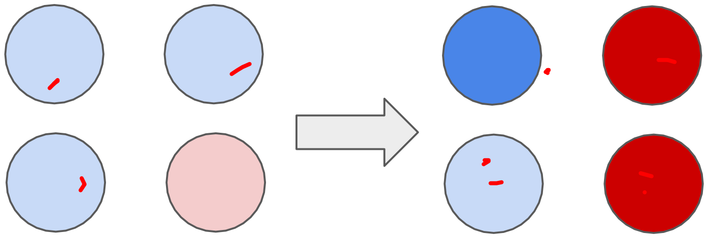
$$\{(D_i, X_i, Y_i)\}_{i=1}^n \xrightarrow{p_{\hat{X},\hat{Y}|X,Y,D}} \{(D_i, \hat{X}_i, \hat{Y}_i)\}_{i=1}^n$$

Calmon el al, 2017

Resampling and Transforming



Resampling



Transforming

Constraint 1: Utility Preservations

- A Utility Function to Preserve the Joint Probability
  - e.g. KL Divergence

$$\sum p(x_i) \log \frac{p(x_i)}{p(x_i)}$$

$$p_{\hat{X},\hat{Y}} \Longleftrightarrow p_{X,Y}$$

transformed data　　　　　　original data

Constraint 2: Discrimination Control

- Constrain the dependency of the target variable y given sensitive feature d to match target $p_{Y_T}(y)$ $\sim 0.1/0.2$
  - J - distance measure $J(p,q) = \left|\frac{p}{q} - 1\right|$
  - $\epsilon_{y,d}$ - a small number used as our tolerance

0.5 0.6 $J(0.5, 0.6) = \left|\frac{0.5}{0.6} - 1\right| \cdot \hat{\sim} 0.1$

$$J\left(p_{\hat{Y}|D}(y|d), p_{Y_T}(y)\right) \leq \epsilon_{y,d} \; \forall \; d \in \mathcal{D}, y \in \{0, 1\}$$

When $p_{\hat{Y}|D}(y|d) = p_{Y_T}(y)$, we achieve Demographic Parity

Constraint 3: Distortion Control

- An Implementation of the Individual Fairness

$$M(x_i) \approx M(x_j) | d(x_i, x_j) \approx 0$$

- The Mapped Sample $\hat{X}, \hat{Y}$ Has to Stay Close to the Original Sample x, y
  - $c_{d,x,y}$ - tolerance
  - δ - a similarity function
    - 1 - very different
    - 0 - very similar

$$\Pr\left(\delta((x, y), (\hat{X}, \hat{Y})) = 1 \mid D = d, X = x, Y = y\right) \leq c_{d,x,y}$$

Putting Things Together

Utility

Discrimination control
group fairness

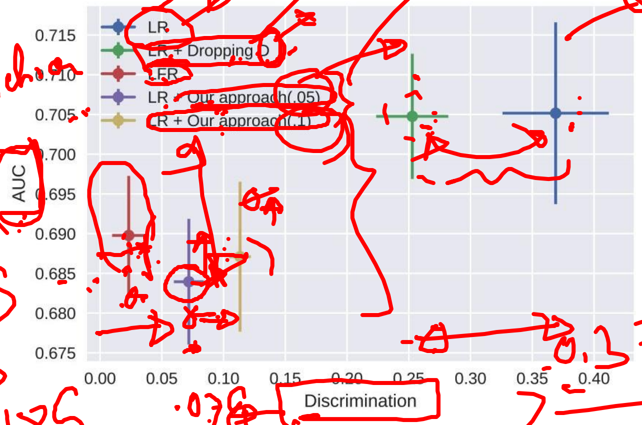
$$
\begin{aligned}
\min_{p_{\hat{X},\hat{Y}|X,Y,D}} \quad & \Delta\left(p_{\hat{X},\hat{Y}}, p_{X,Y}\right) \\
\text{s.t.} \quad & J\left(p_{\hat{Y}|D}(y|d), p_{Y_T}(y)\right) \leq \epsilon_{y,d} \ \text{ and} \\
& \mathbb{E}\left[\underbrace{\delta((x,y),(\hat{X},\hat{Y}))}_{\text{Individual}} \mid D=d, X=x, Y=y\right] \leq c_{d,x,y}
\end{aligned}
$$

"DP"

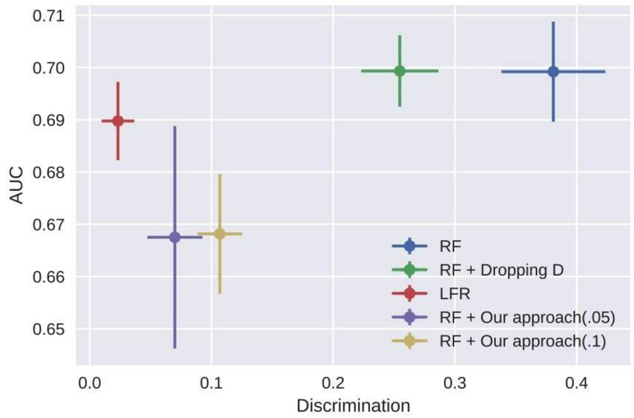
Distortion Control
Individual fairness

Results on COMPAS dataset

ROC



Logistic Regression



Random Forest

LFR - Learning Fair Representations (benchmark)

Prejudice Remover Regularizer
Quantified causes of unfairness

Prejudice

– Unfairness rooted in the dataset

Underestimation

– Model unfairness because the model is not fully converged

Negative Legacy

– Unfairness due to sampling biases

Training Objective

D + parametes

θ + $(\theta_1^2 + \theta_2^2 + \theta_3^2 + \ldots)$

$$-\mathcal{L}(\mathcal{D};\boldsymbol{\Theta}) + \eta\,\mathrm{R}(\mathcal{D},\boldsymbol{\Theta}) + \frac{\lambda}{2}\|\boldsymbol{\Theta}\|_2^2$$

Loss of the Model | Fairness Regularizer | L2 Regularizer

[Kamishima et al, 2012](#)

Prejudice Index (PI)

Recall that Indirect Discrimination Happens When

Prediction is not directly conditioned on sensitive variables S

Prediction is indirectly conditioned on S by a variable O that is dependent on S

Prejudice Index (PI)

- Measures the degree of indirect discrimination based on mutual information

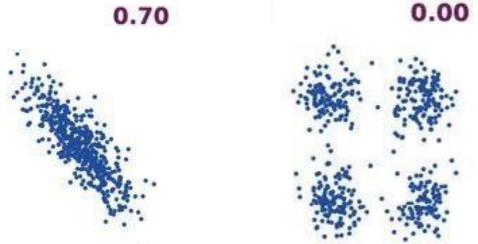
output sensitive

$$\mathrm{PI} = \sum_{(y,s)\in\mathcal{D}} \hat{\Pr}[y, s] \ln \frac{\hat{\Pr}[y, s]}{\hat{\Pr}[y]\hat{\Pr}[s]}$$

prediction model

1

$\downarrow$ PI ≈ 0



some prejudice little prejudice

Kamishima et al, 2012

Normalized Prejudice Index (NPI)

- Prejudice Index (PI)
  - Measures the degree of indirect discrimination based on mutual information
  - Ranges in [0, +∞)

$$\mathrm{PI} = \sum_{(y,s)\in\mathcal{D}} \hat{\Pr}[y,s] \ln \frac{\hat{\Pr}[y,s]}{\hat{\Pr}[y]\hat{\Pr}[s]}$$

$\sum y_i \ln y_i$

- Normalized Prejudice Index (NPI)
  - Normalize PI by the entropy of Y and S
  - Ranges in [0, 1]

$$\mathrm{NPI} = \mathrm{PI}/(\sqrt{\mathrm{H}(Y)\mathrm{H}(S)})$$

entropy

Kamishima et al, 2012

Optimizing PI

- Learning PI

NPL

$$\mathrm{PI} = \sum_{Y,S} \hat{\Pr}[Y,S] \ln \frac{\hat{\Pr}[Y,S]}{\hat{\Pr}[S]\hat{\Pr}[Y]}$$

Kamishima et al, 2012

Optimizing PI

$Pr(Y,S) = \sum_{x} Pr(X,Y,S)$

features

- Learning PI

Expands Pr(Y, S) into Σ_xPr(X, Y, S)

$$\mathrm{PI} = \sum_{Y,S} \hat{\Pr}[Y,S] \ln \frac{\hat{\Pr}[Y,S]}{\hat{\Pr}[S]\hat{\Pr}[Y]} = \sum_{X,S} \tilde{\Pr}[X,S] \sum_{Y} \mathcal{M}[Y|X,S;\Theta] \ln \frac{\hat{\Pr}[Y,S]}{\hat{\Pr}[S]\hat{\Pr}[Y]}.$$

double summations

triple summations

Prediction Model

- Using Logistic Regression Model as the Prediction Model

$$\mathcal{M}[y|\mathbf{x},s;\Theta] = y\sigma(\mathbf{x}^\top \mathbf{w}_s) + (1-y)(1-\sigma(\mathbf{x}^\top \mathbf{w}_s))$$

Kamishima et al, 2012

Optimizing PI

$P(Y,Y) = \sum P(X,Y)$

Y is a fn. of both X, S

- Learning PI

$$\mathrm{PI} = \sum_{Y,S} \hat{\Pr}[Y,S] \ln \frac{\hat{\Pr}[Y,S]}{\hat{\Pr}[S]\hat{\Pr}[Y]} = \sum_{X,S} \tilde{\Pr}[X,S] \sum_{Y} \mathcal{M}[Y|X,S;\boldsymbol{\Theta}] \ln \frac{\hat{\Pr}[Y,S]}{\hat{\Pr}[S]\hat{\Pr}[Y]}.$$

$P(Y,S) = \sum P(X,Y,S)$

$$= \sum_{(\mathbf{x}_i,s_i)\in\mathcal{D}} \sum_{y\in\{0,1\}} \mathcal{M}[y|\mathbf{x}_i,s_i;\boldsymbol{\Theta}] \ln \boxed{\frac{\hat{\Pr}[y|s_i]}{\hat{\Pr}[y]}}.$$

$s_i = 0$

difficult to evaluate

- Using Logistic Regression Model as the Prediction Model

$$\mathcal{M}[y|\mathbf{x},s;\boldsymbol{\Theta}] = y\sigma(\mathbf{x}^\top \mathbf{w}_s) + (1-y)(1-\sigma(\mathbf{x}^\top \mathbf{w}_s))$$

[Kamishima et al, 2012](#)

Optimizing PI

$$\mathrm{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\Pr}[y|s_i]}{\hat{\Pr}[y]}$$

Kamishima et al, 2012

Optimizing PI

$$\mathrm{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\boxed{\hat{\Pr}[y|s_i]}}{\hat{\Pr}[y]}$$

$$\hat{\Pr}[y|s] = \int_{\mathrm{dom}(X)} \Pr{}^*[X|s]\mathcal{M}[y|X, s; \boldsymbol{\Theta}]dX$$

Integrals Are Difficult to Evaluate

Kamishima et al, 2012

Optimizing PI

$$\mathrm{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\boxed{\hat{\Pr}[y|s_i]}}{\hat{\Pr}[y]}$$

$$\hat{\Pr}[y|s] = \int_{\mathrm{dom}(X)} \Pr{}^*[X|s] \mathcal{M}[y|X, s; \boldsymbol{\Theta}] dX$$

$$\approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s} \mathcal{M}[y|\mathbf{x}_i, s; \boldsymbol{\Theta}]}{|\{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s\}|}$$

Approximating integrals by sample means

Kamishima et al, 2012

Optimizing PI

$$\mathrm{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\boxed{\hat{\Pr}[y|s_i]}}{\hat{\Pr}[y]}$$

$$\hat{\Pr}[y|s] = \int_{\mathrm{dom}(X)} \Pr{}^*[X|s] \mathcal{M}[y|X, s; \boldsymbol{\Theta}] dX$$

$$\approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s} \mathcal{M}[y|\mathbf{x}_i, s; \boldsymbol{\Theta}]}{|\{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s\}|}$$

Approximating integrals by sample means

Kamishima et al, 2012

Optimizing PI

$S_i = 0$, $S_i : 0$

$$\mathrm{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\Pr}[y|s_i]}{\boxed{\hat{\Pr}[y]}}$$

$$\hat{\Pr}[y|s] = \int_{\mathrm{dom}(X)} \Pr{}^*[X|s] \mathcal{M}[y|X, s; \boldsymbol{\Theta}] dX$$

$$\approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s} \mathcal{M}[y|\mathbf{x}_i, s; \boldsymbol{\Theta}]}{|\{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s\}|}$$

$$\hat{\Pr}[y] \approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}]}{|\mathcal{D}|}$$

Approximating integrals by sample means

Kamishima et al, 2012

Optimizing PI

$\Pr[y|0]$
$\Pr[y|1]$

$$\mathrm{PI} = \sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \sum_{y \in \{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\Pr}[y|s_i]}{\hat{\Pr}[y]}$$

$\dfrac{\Pr[y|0]}{P(y)}$

$\dfrac{\Pr[y|1]}{P(y)}$

$$\hat{\Pr}[y|\boxed{s}] \approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s} \mathcal{M}[y|\mathbf{x}_i, s; \boldsymbol{\Theta}]}{|\{(\mathbf{x}_i, s_i) \in \mathcal{D} \text{ s.t. } s_i = s\}|} \qquad \hat{\Pr}[y] \approx \frac{\sum_{(\mathbf{x}_i, s_i) \in \mathcal{D}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}]}{|\mathcal{D}|}$$

$\hat{\Pr}(y)$

[Kamishima et al, 2012]

Putting Things Together

Optimization Target

$$-\mathcal{L}(\mathcal{D};\boldsymbol{\Theta}) + \eta \mathrm{R}(\mathcal{D},\boldsymbol{\Theta}) + \frac{\lambda}{2}\|\boldsymbol{\Theta}\|_2^2$$

Loss of the Model　　Fairness Regularizer　　L2 Regularizer

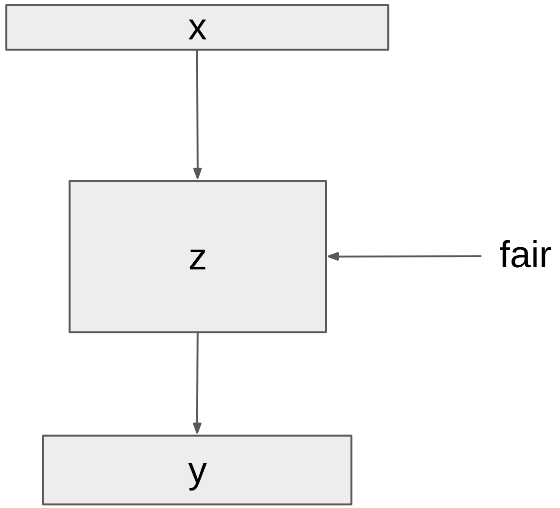
- Fairness Regularizer

$$\mathrm{PI} = \sum_{(\mathbf{x}_i, s_i)\in\mathcal{D}} \sum_{y\in\{0,1\}} \mathcal{M}[y|\mathbf{x}_i, s_i; \boldsymbol{\Theta}] \ln \frac{\hat{\Pr}[y|s_i]}{\hat{\Pr}[y]}$$

Kamishima et al, 2012

Fair Representation Learning

– Ensure fairness up to a certain level

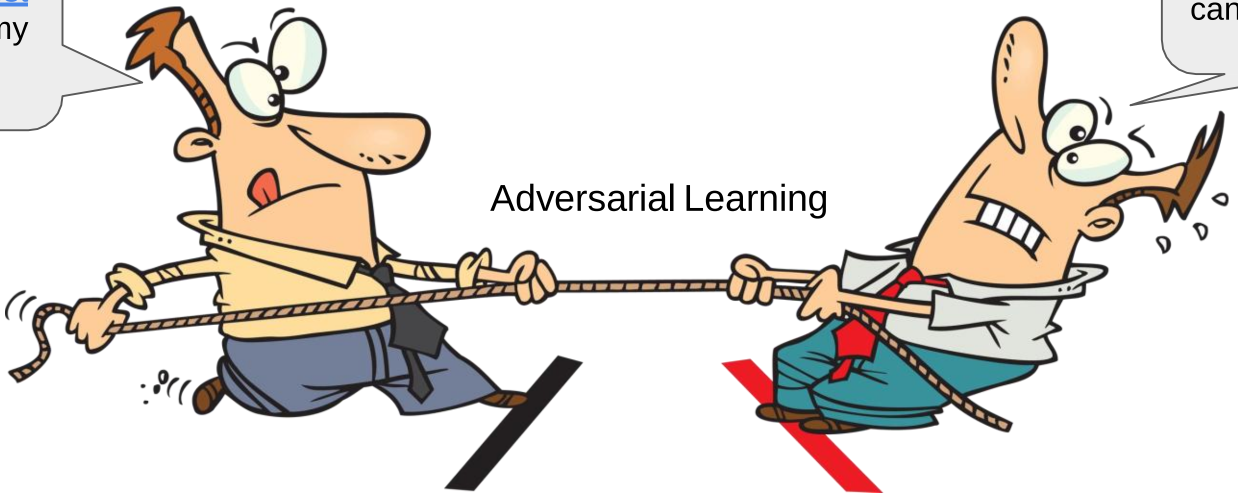


Fair Representation Learning

- How Do We Test the Fairness of Representation Z?
  - Adversarial Learning

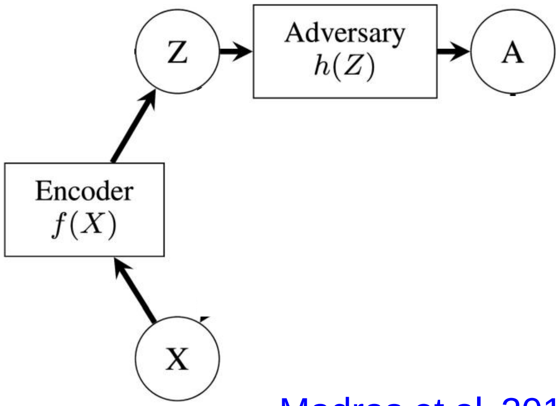
I want to find the best representation for my task.

I want to find the worst representation that can reconstruct A



Fair Representations

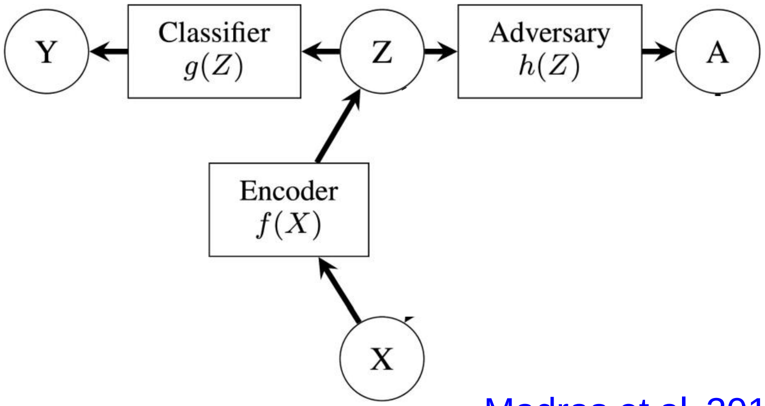
- How Do We Make a Representation Z Fair?
  - Z = f(X)
  - Test and see if a good amount of A can be reconstructed from Z
  - Compare A with h(z)



Madras et al, 2018

Fair Representations

- How Do We Make a Representation Z Fair?
  - Z = f(X)
  - Test and see if a good amount of A can be reconstructed from Z
  - Compare A with h(z)
- Properties of Deep Representations
  - Achieve good performance for downstream task that generates y=g(z)

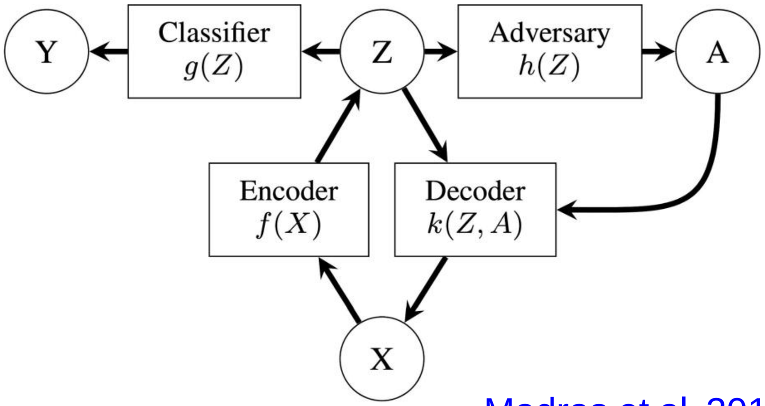


Madras et al, 2018

Fair Representations

- How Do We Make a Representation Z Fair?
  - Z = f(X)
  - Test and see if a good amount of A can be reconstructed from Z
  - Compare A with h(z)

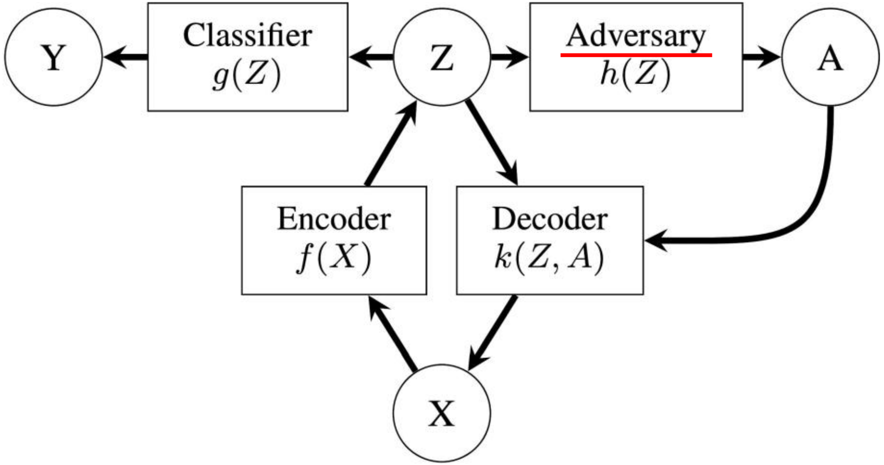
- Properties of Deep Representations
  - Achieve good performance for downstream task that generates y=g(z)
  - Has the Ability to Reconstruct X = k(Z,A)



Madras et al, 2018

Fairness Through Adversarial Learning

- Adversarial Learning
    - Models are trained using objectives that compete with each other



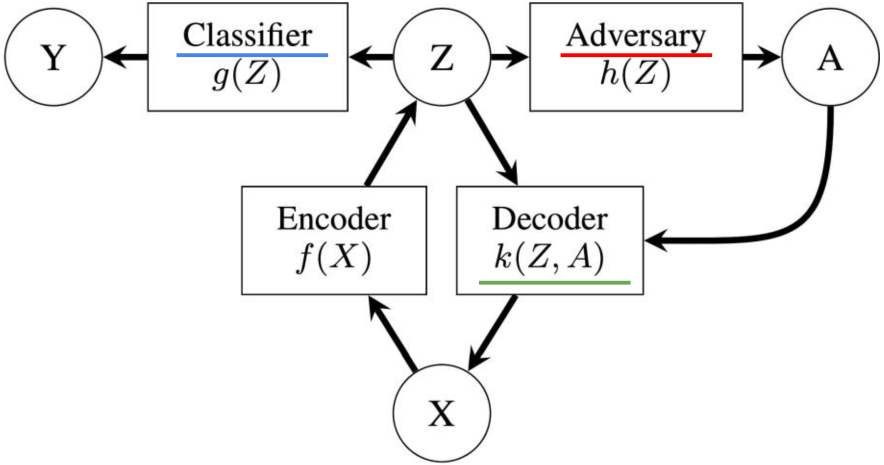
$$\underset{f,g,k}{\text{minimize}}\ \underset{h}{\text{maximize}}\ \mathbb{E}_{X,Y,A}\left[L(f,g,h,k)\right]$$

Madras et al, 2018

Fairness Through Adversarial Learning

- Adversarial Learning

$$L(f,g,h,k) = \underline{\alpha L_C(g(f(X,A)),Y)} + \underline{\beta L_{Dec}(k(f(X,A),A),X)} + \gamma \underline{L_{Adv}(h(f(X,A)),A)}$$



$$\underset{f,g,k}{\text{minimize}}\ \underset{h}{\text{maximize}}\ \mathbb{E}_{X,Y,A}\left[L(f,g,h,k)\right]$$

[Madras et al, 2018]

Loss for Learning Fair Representations

- Adversarial Loss for Demographic Parity with Group $\mathcal{D}_0, \mathcal{D}_1$

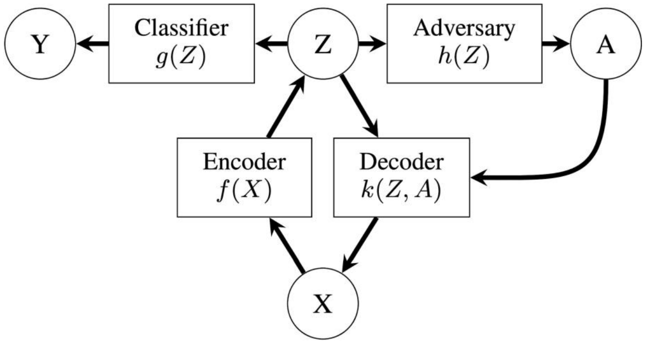
$$L_{Adv}^{DP}(h) = 1 - \sum_{i \in \{0,1\}} \frac{1}{|\mathcal{D}_i|} \sum_{(x,a) \in \mathcal{D}_i} |h(f(x,a)) - a|$$

Demographic Parity: $P(\hat{Y} = 1 | A = 1) = P(\hat{Y} = 1 | A = 0)$

- Adversarial Loss for Equality of Odds with Group $\mathcal{D}_i^j = \{(x, y, a) \in \mathcal{D} | a = i, y = j\}$

$$L_{Adv}^{EO}(h) = 2 - \sum_{(i,j) \in \{0,1\}^2} \frac{1}{|\mathcal{D}_i^j|} \sum_{(x,a) \in \mathcal{D}_i^j} |h(f(x,a)) - a|$$

Equality of Odds: $P(\hat{Y} = 1 | A = 0, Y) = P(\hat{Y} = 1 | A = 1, Y)$



Madras et al, 2018

Discrimination Measures for Representations

$$\mathcal{Z}_1 = p(Z|A=1) \quad \mathcal{Z}_0 = p(Z|A=0) \quad \mathcal{Z}_a^y = p(Z|A=a, Y=y)$$

- Demographic Parity

$$\Delta_{DP}(g) \triangleq d_g(\mathcal{Z}_0, \mathcal{Z}_1) = |\mathbb{E}_{\mathcal{Z}_0}[g] - \mathbb{E}_{\mathcal{Z}_1}[g]|$$

Demographic Parity: $P(\hat{Y}=1|A=1) = P(\hat{Y}=1|A=0)$

- Equality of Odds

$$\Delta_{EO}(g) \triangleq |\mathbb{E}_{\mathcal{Z}_0^0}[g] - \mathbb{E}_{\mathcal{Z}_1^0}[g]| + |\mathbb{E}_{\mathcal{Z}_0^1}[1-g] - \mathbb{E}_{\mathcal{Z}_1^1}[1-g]|$$

Equality of Odds: $P(\hat{Y}=1|A=0,Y) = P(\hat{Y}=1|A=1,Y)$

- Equality of Opportunities

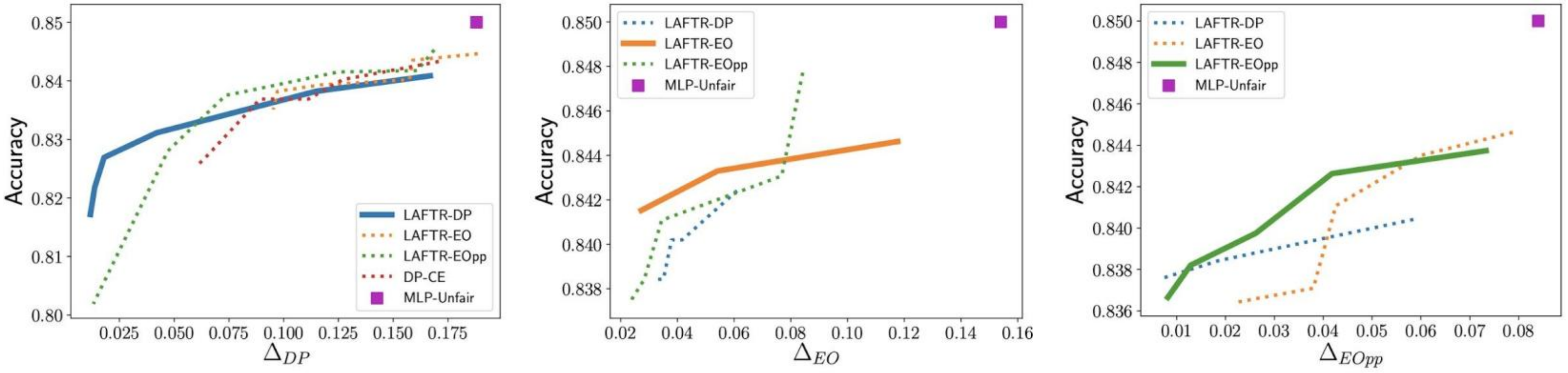
$$\Delta_{EOpp}(g) \triangleq |\mathbb{E}_{\mathcal{Z}_0^0}[g] - \mathbb{E}_{\mathcal{Z}_1^0}[g]|$$

Equality of Opportunity: $P(\hat{Y}=1|A=0,Y=1) = P(\hat{Y}=1|A=1,Y=1)$

Accuracy and Fairness on Adult Income Dataset

- Results Generated By Varying γ.

$$L(f,g,h,k) = \alpha L_C(g(f(X,A)),Y) + \beta L_{Dec}(k(f(X,A),A),X) + \gamma L_{Adv}(h(f(X,A)),A)$$

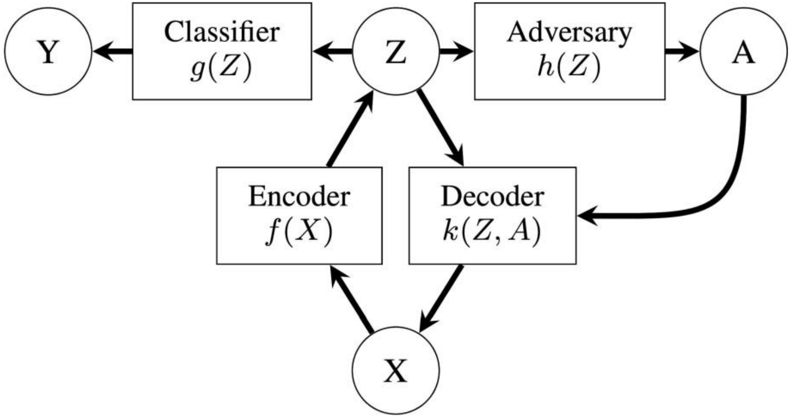


Madras et al, 2018

DP-CE - Cross Entropy Adversarial Objective (Edwards et al, 2016)

Transferring Fair Representations

- If the Representations Are Fair, All Predictors Should Be Fair!
  - Train f and g based on domain 1 with feature space X
  - Fix f, and train g' on domain 2 with the same feature space X
    - y=g'(f(x)) should be a fairness predictor



Madras et al, 2018

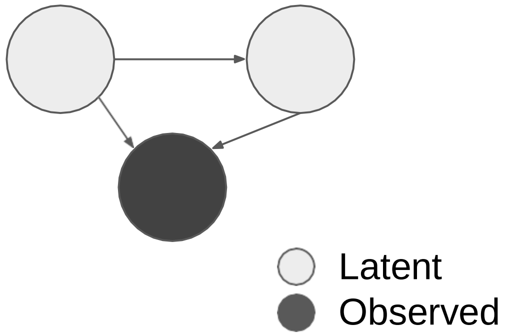
Comparisons: Regularization and Adversarial Learning

	Prejudice Removing Regularizer	Adversarial Learning
Pros	Minimal modifications to training procedure	Transferable representations
		Can be applied to many different fairness criteria
Cons	Can only be applied to Demographic Parity	Adversarial loss can be difficult to train

Fair Causal Reasoning

Causal Graph

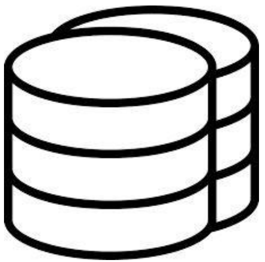
- Observed Data
- Latent Data
- Relations



Causal Fairness Criteria

- Counterfactual Fairness

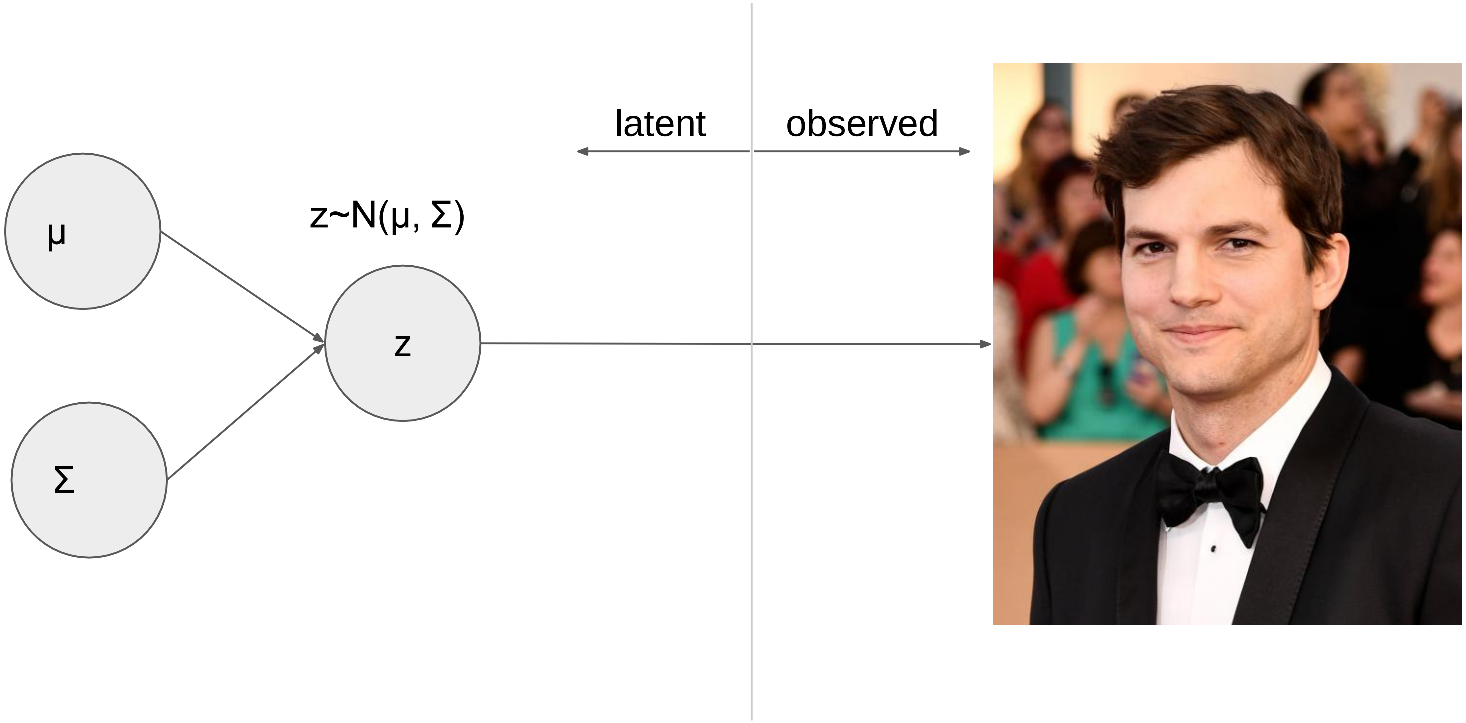
Observed Data



Observational Fairness Criteria

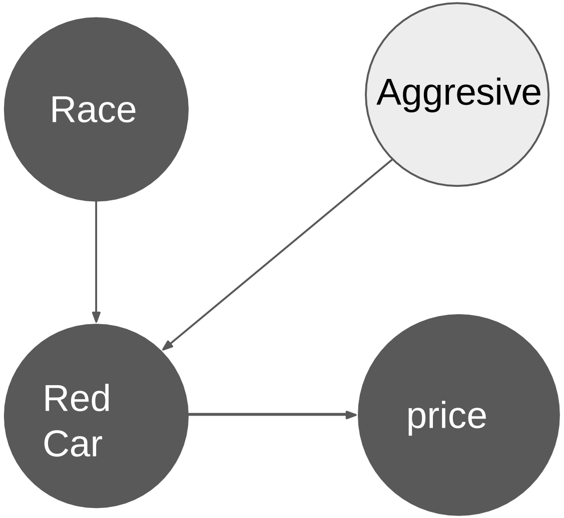
- Fairness Through Unawareness
- Demographic Parity
- Equalized Odds/Opp

Causal Graph



Why Do We Need Causal Fairness?

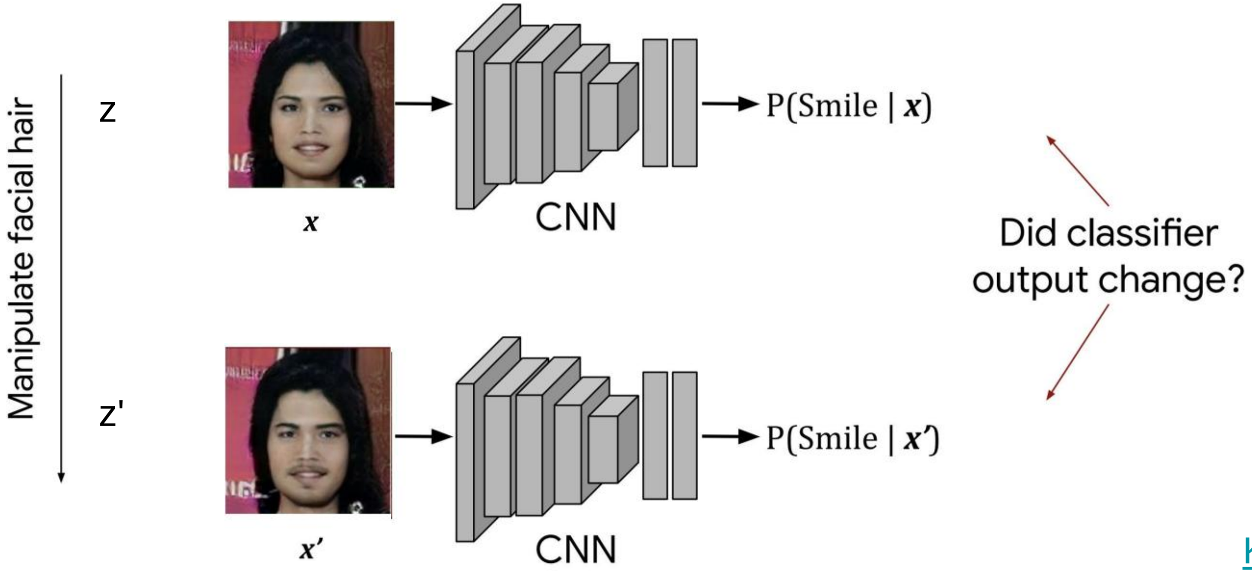
- Recover Latent Variables



Kusner et al, 2018

Why Do We Need Causal Fairness?

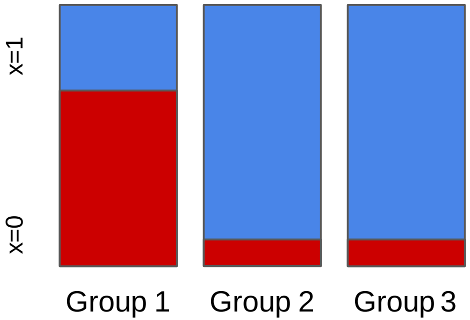
- Recover Latent Variables



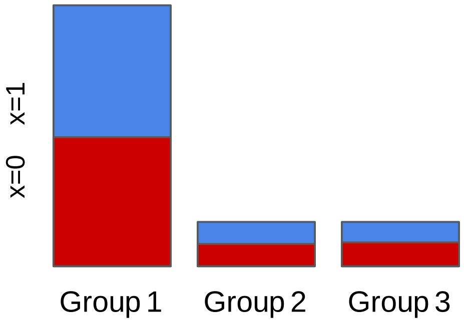
Kusner et al, 2018

Why Do We Need Causal Fairness?

- Dealing with Inherent bias



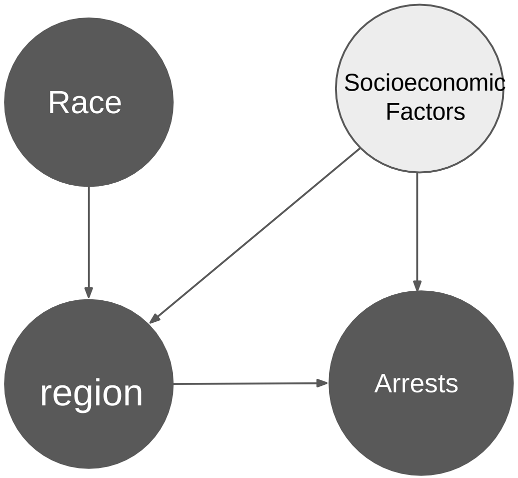
Inherent Biases



Sampling Biases

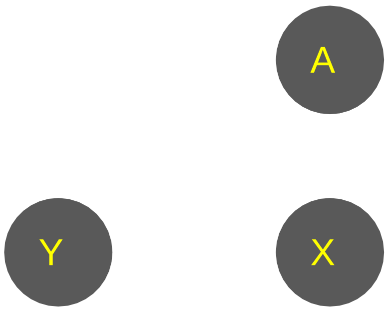
Inherent bias

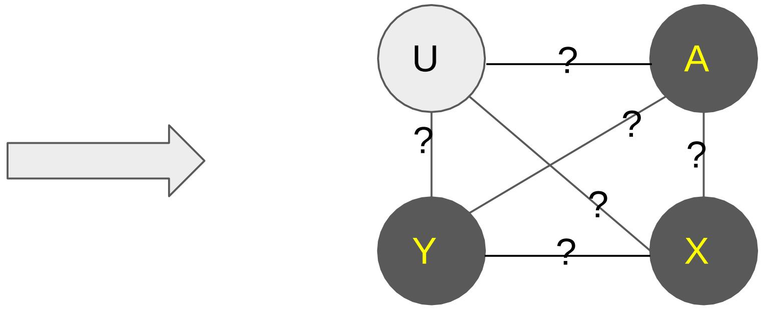
- Race groups live in certain regions due to socioeconomic status
- Latent Socioeconomic factors
  - More police resources in regions with low economic status
  - Results in more arrests



Kusner et al, 2018

Counterfactual Fairness

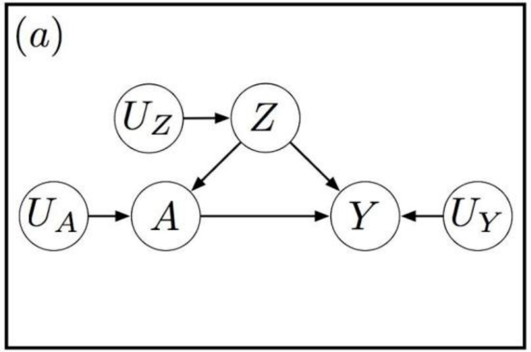




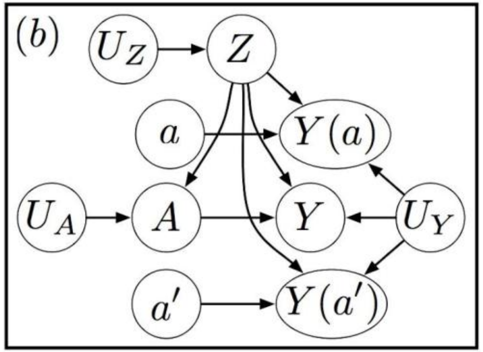
$$P(\hat{Y}_{A\leftarrow a}(U) = y \mid X = x, A = a) = P(\hat{Y}_{A\leftarrow a'}(U) = y \mid X = x, A = a)$$

Real Examples	Counterfactual Examples
Intervention on A←a	Intervention on A←a'

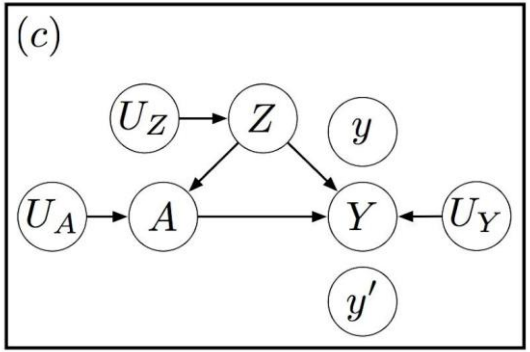
Intervention on Causal Graphs



Causal Graph with A, Z, Y



Intervene on A

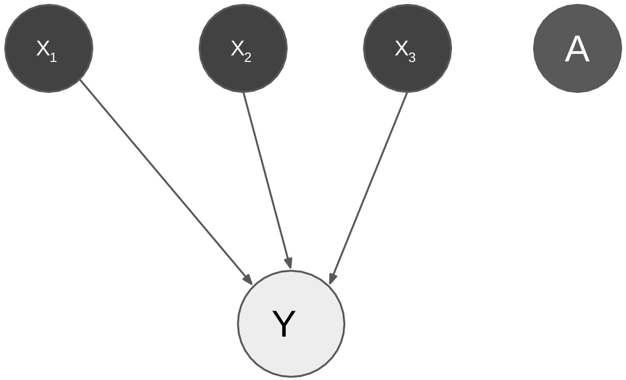


Intervene on Y

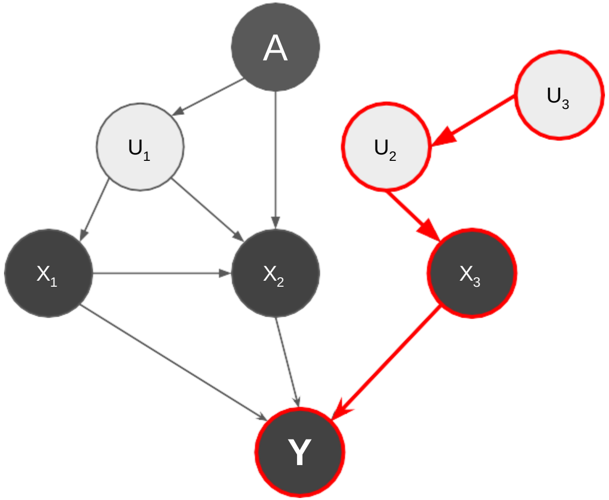
[Loftus et al, 2018]

Counterfactual Fairness

- Level 1
  - Build predictors using only the observable non-descendants of A

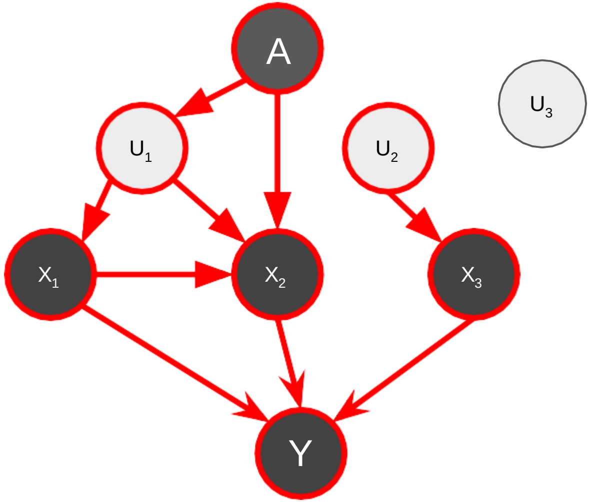


Fairness Through Unawareness



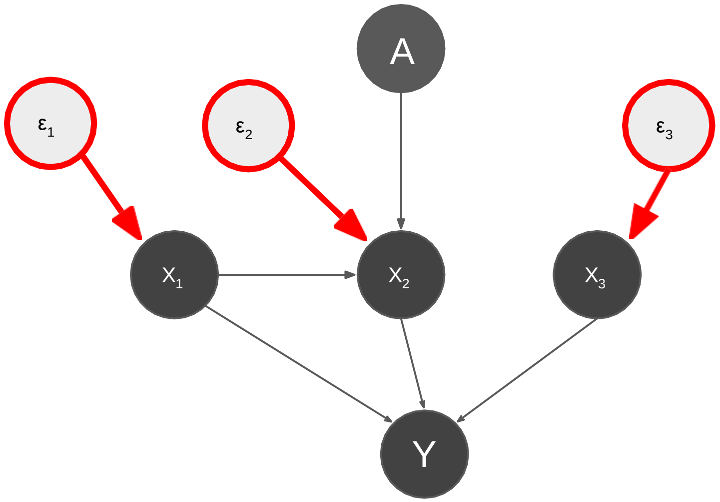
Counterfactual Fairness

- Level 2
  - Build Predictors using the parents of the observable variables



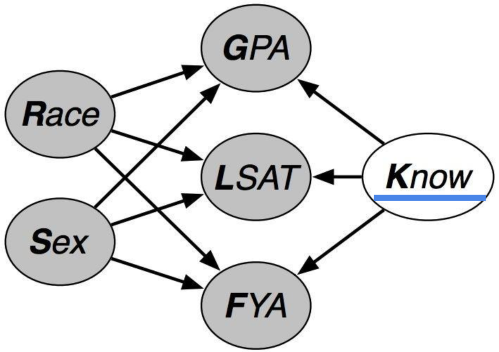
Counterfactual Fairness

- Level 3
  - Build Predictors by adding independent error terms



Level 2 Counterfactual Fairness

- Build Predictors using the parents of the observable variables



$$K \sim \mathcal{N}(0, 1)$$

$$\text{FYA} \sim \mathcal{N}(w_F^K K + w_F^R R + w_F^S S, 1) \qquad \text{GPA} \sim \mathcal{N}(b_G + w_G^K K + w_G^R R + w_G^S S, \sigma_G)$$

Gaussian Dist.

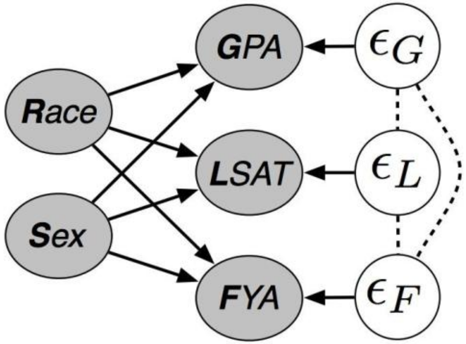
$$\text{LSAT} \sim \text{Poisson}(\exp(b_L + w_L^K K + w_L^R R + w_L^S S))$$

Parameters

[Kusner et al, 2018]

Level 3 Counterfactual Fairness

- Build Predictors by adding independent error terms

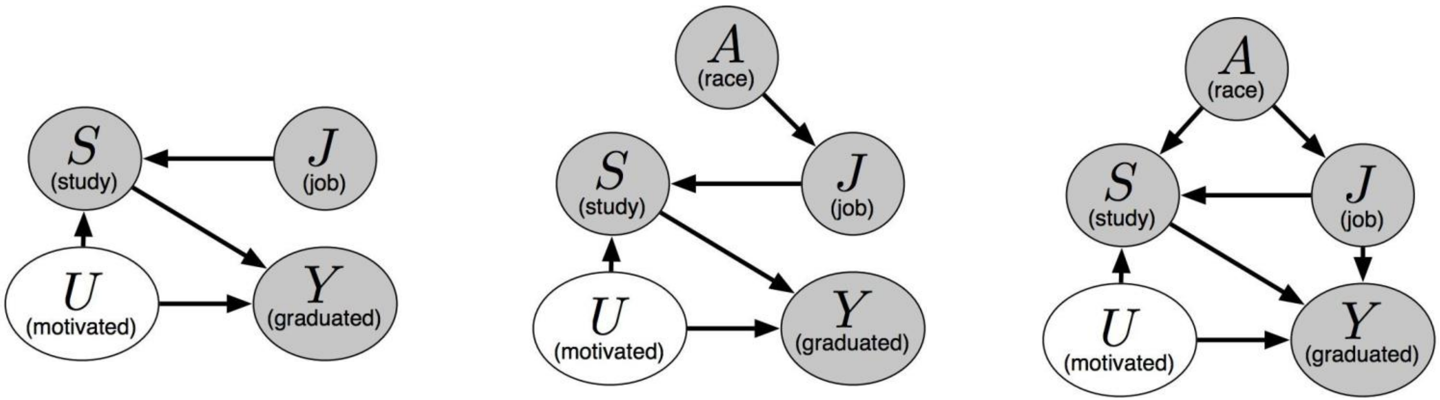


$$\text{GPA} = b_G + w_G^R R + w_G^S S + \epsilon_G, \quad \epsilon_G \sim p(\epsilon_G)$$
$$\text{LSAT} = b_L + w_L^R R + w_L^S S + \epsilon_L, \quad \epsilon_L \sim p(\epsilon_L)$$
$$\text{FYA} = b_F + w_F^R R + w_F^S S + \epsilon_F, \quad \epsilon_F \sim p(\epsilon_F)$$

Kusner et al, 2018

Multiple Causal Graphs

- Whether a student can graduate on time



[Russell et al, 2017](#)

Alternative Definitions of Counterfactual Fairness

Exact Formulation

$$P(\hat{Y}_{A\leftarrow a}(U) = y \mid X = x, A = a) = P(\hat{Y}_{A\leftarrow a'}(U) = y \mid X = x, A = a)$$

ε - Approximate Formulation

$$\left| f(\mathbf{x}_{A\leftarrow a}, a) - f(\mathbf{x}_{A\leftarrow a'}, a') \right| \leq \epsilon$$

(δ, ε) - Approximate Formulation

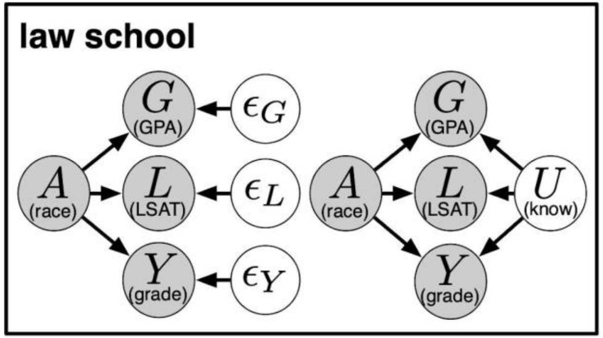
$$\mathbb{P}(\left| f(\mathcal{X}_{A\leftarrow a}, a) - f(\mathcal{X}_{A\leftarrow a'}, a') \right| \leq \epsilon \mid \mathcal{X} = \mathbf{x}, A = a) \geq 1 - \delta$$

Multi-world Counterfactual Fairness

$$\min_f \frac{1}{n}\sum_{i=1}^{n} \underbrace{\ell(f(\mathbf{x}_i, a_i), y_i)}_{\text{loss of the data}} + \lambda \underbrace{\sum_{j=1}^{m}}_{\text{world j}} \frac{1}{n}\sum_{i=1}^{n} \sum_{a' \neq a_i} \mu_j(f, \underbrace{\mathbf{x}_i}_{\text{counterfactual examples}}, a_i, a')$$

$$\mu_j(f, \mathbf{x}_i, a_i, a') := \underbrace{\frac{1}{S}\sum_{s=1}^{S}}_{\text{Monte-carlo Samples}} \max\{0, \underbrace{\left| f(\mathbf{x}^{s}_{i, A^j \leftarrow a_i}, a_i) - f(\mathbf{x}^{s}_{i, A^j \leftarrow a'}, a') \right| - \epsilon}_{\text{ε - Approximate Counterfactual Fairness}}\}$$

Law Graduate School



$$G = b_G + w_G^A A + \epsilon_G$$
$$L = b_L + w_L^A A + \epsilon_L$$
$$Y = b_Y + w_Y^A A + \epsilon_Y$$
$$\epsilon_G, \epsilon_L, \epsilon_Y \sim \mathcal{N}(0,1)$$

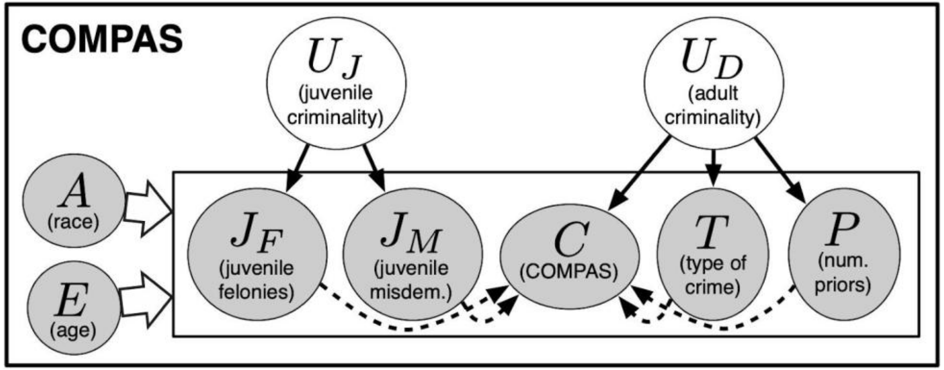
L3 Method

$$G \sim \mathcal{N}(b_G + w_G^A A + w_G^U U, \sigma_G)$$
$$L \sim \text{Poisson}(\exp(b_L + w_L^A A + w_L^U U))$$
$$Y \sim \mathcal{N}(w_Y^A A + w_Y^U U, 1)$$
$$U \sim \mathcal{N}(0,1)$$

L2 Method

Russell et al, 2017

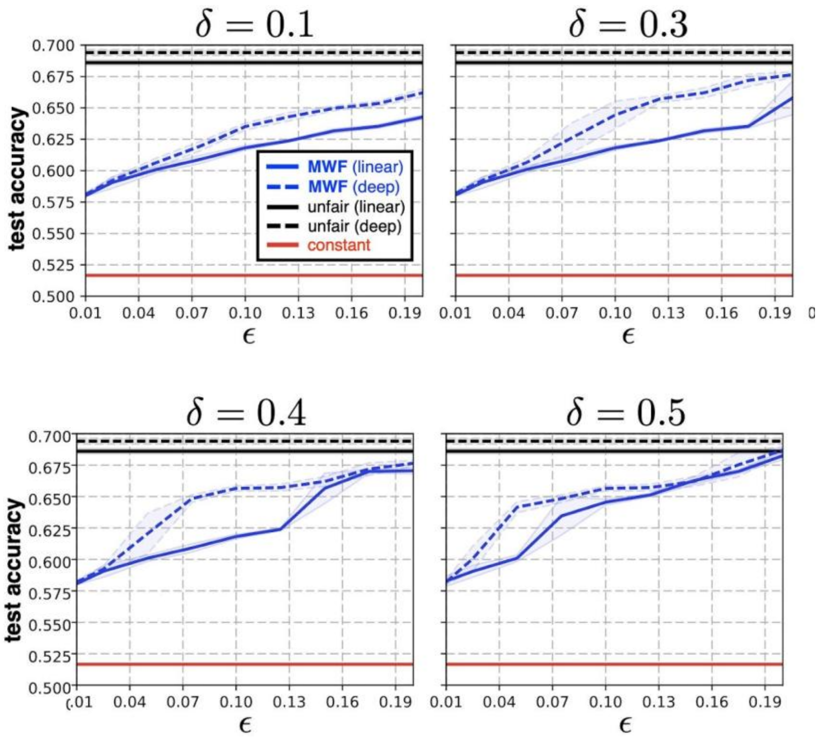
COMPAS



$$
\begin{aligned}
T &\sim \text{Bernoulli}(\phi(b_T + w_C^{U_D} U_D + w_C^E E + w_C^A A) \quad (\\
C &\sim \mathcal{N}(b_C + w_C^{U_D} U_D + w_C^E E + w_C^A A + w_C^T T + w_C^P P + w_C^{J_F} J_F + w_C^{J_M} J_M, \sigma_C) \\
P &\sim \text{Poisson}(\exp(b_P + w_P^{U_D} U_D + w_P^E E + w_P^A A)) \\
J_F &\sim \text{Poisson}(\exp(b_{J_F} + w_{J_F}^{U_J} + w_{J_F}^E E + w_{J_F}^A A)) \\
J_M &\sim \text{Poisson}(\exp(b_{J_M} + w_{J_M}^{U_J} + w_{J_M}^E E + w_{J_M}^A A)) \\
[U_J, U_D] &\sim \mathcal{N}(0, \Sigma)
\end{aligned}
$$

Russell et al, 2017

Results



Russell et al, 2017

Equalized Counterfactual Odds

Equality of Odds

$$P(\hat{Y}=1|A=0,Y)=P(\hat{Y}=1|A=1,Y)$$

Counterfactual Fairness

$$P(\underline{\hat{Y}_{A\leftarrow a}\,(U)}=y\mid X=x,A=a)=P(\underline{\hat{Y}_{A\leftarrow a'}(U)}=y\mid X=x,A=a)$$

Equalized Counterfactual Odds

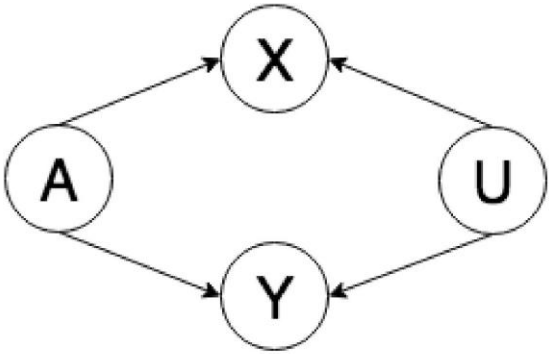
$$p(\underline{\hat{Y}_{A\leftarrow a}(U)}\mid X=x,\underline{Y_{A\leftarrow a}}=y,A=a)=p(\underline{\hat{Y}_{A\leftarrow a'}(U)}\mid X=x,\underline{Y_{A\leftarrow a'}}=y,A=a)$$

- *"the predictor is counterfactually fair, conditioned on the factual outcome Y matching the counterfactual outcome $Y_{A\leftarrow a}$.*
- *In contrast, the original counterfactual fairness formulation can be interpreted as requiring predictions to be the same across factual counterfactual pairs, regardless of whether those pairs share the same value of the outcome."*

Healthcare Equality

- **Goal:** *The same prediction to be made for a patient, and a counterfactual patient resulting from changing a sensitive attribute*
  - if the factual and counterfactual outcomes do not differ.

- Protected Features A = {Gender}
- Features X, vector representation of coded diagnoses, procedures, medication orders, lab results, and clinical notes
- Prediction Y, a binary indicator of the occurrence of a clinically relevant outcome



$$u \sim p(U) = \mathrm{Normal}(0, I)$$
$$a \sim p(A) = \mathrm{Categorical}(A \mid \pi)$$
$$x, y \sim p(X, Y \mid U, A) = p(X \mid U, A)p(Y \mid U, A)$$

https://arxiv.org/pdf/1907.06260

Training Objective

- σ - sigmoid function
- h - predictor
- J - cross entropy loss

$$\mathcal{L} = J(h_\theta(x,a),y) + \lambda_{\text{CF}} \sum_{a_k \in \mathcal{A}} \mathbb{1}[a \neq a_k] J(h_\theta(\underline{x_{A\leftarrow a_k}}, a_k), \underline{y_{A\leftarrow a_k}}) +$$

$$\lambda_{\text{CLP}} \sum_{a_k \in \mathcal{A}} \mathbb{1}[a \neq a_k] \mathbb{1}[y = \underline{y_{A\leftarrow a_k}}] \Big(\underbrace{\sigma^{-1}(h_\theta(\underline{x_{A\leftarrow a_k}}, a_k))}_{\text{logits}} - \underbrace{\sigma^{-1}(h_\theta(x,a))}_{\text{logits}} \Big)^2$$

- Counterfactual Logit Pairing (CLP) is a technique that addresses such issues to ensure that a model's prediction doesn't change when a sensitive attribute referenced in an example is either removed or replaced. It improves a model's robustness to such perturbations, and can positively influence a model's stability, fairness, and safety.

Fair NLP

- Biases in NLP Models
- Data Augmentation
- Debiasing Word Embedding
- Adversarial Learning

Biases of NLP Models

- Denigration
  - The use of culturally or historically derogatory terms
- Under-representation
  - The disproportionately low representation of a specific group
  - e.g., a classifier's performance is adversely affected due to sampling biases of the minority protected group
- Stereotyping
  - An over-generalized belief about a particular category of people
  - e.g., a classifier attributes man to computers more than woman
- Recognition
  - Algorithms perform different for protected groups because of their inherent characteristics
  - e.g., a voice recognition algorithm has better capabilities in recognizing voices in low frequency

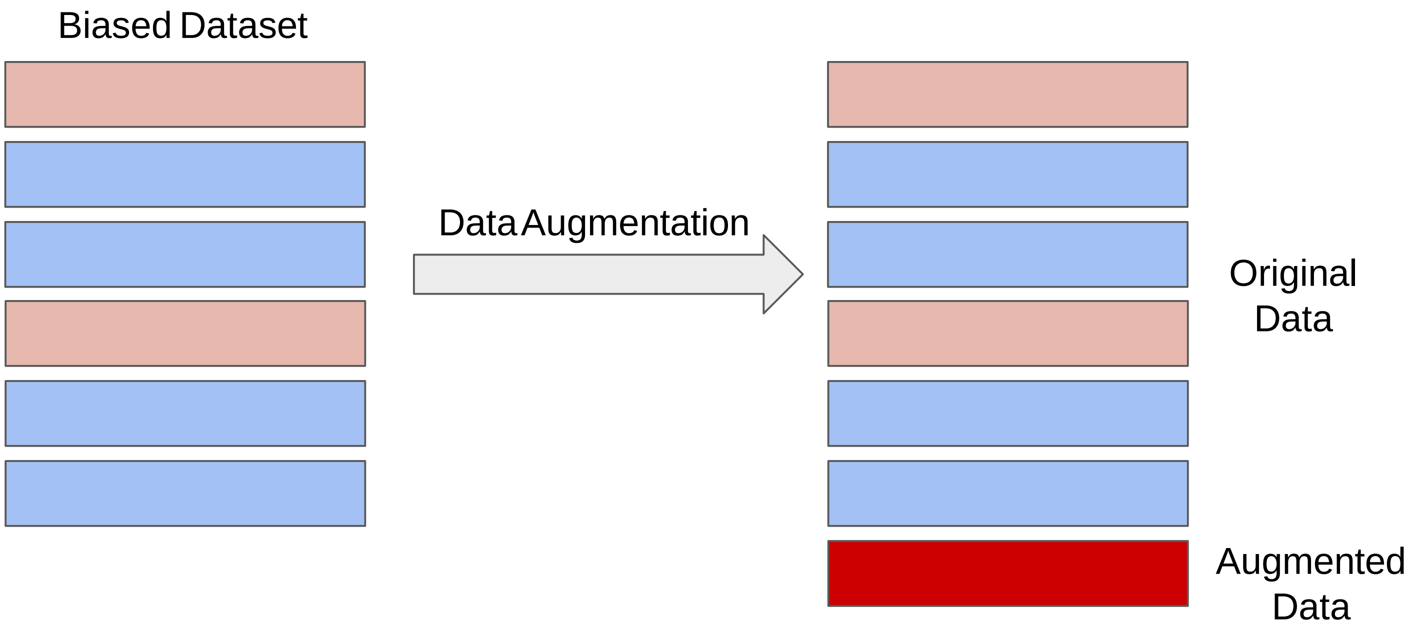
Biases of NLP Models

Task	Example of Representation Bias in the Context of Gender	S
Machine Translation	Translating "He is a nurse. She is a doctor." to Hungarian and back to English results in "She is a nurse. He is a doctor." (Douglas, 2017)	✓
Caption Generation	An image captioning model incorrectly predicts the agent to be male because there is a computer nearby (Burns et al., 2018).	✓
Speech Recognition	Automatic speech detection works better with male voices than female voices (Tatman, 2017).	✗
Sentiment Analysis	Sentiment Analysis Systems rank sentences containing female noun phrases to be indicative of anger more often than sentences containing male noun phrases (Park et al., 2018).	✓
Language Model	"He is doctor" has a higher conditional likelihood than "She is doctor" (Lu et al., 2018).	✓
Word Embedding	Analogies such as "man : woman :: computer programmer : homemaker" are automatically generated by models trained on biased word embeddings (Bolukbasi et al., 2016).	✓

(S)tereotyping, (D)enigration, (R)ecognition, (U)nder-representation

Sun et al, 2019

Data Augmentation



Coreference Resolution

A man and his son get into a terrible car crash. The father dies, and the boy is badly injured. In the hospital, the surgeon looks at the patient and exclaims, “I can’t operate on this boy, he’s my son!

Does this paragraph make sense to you?

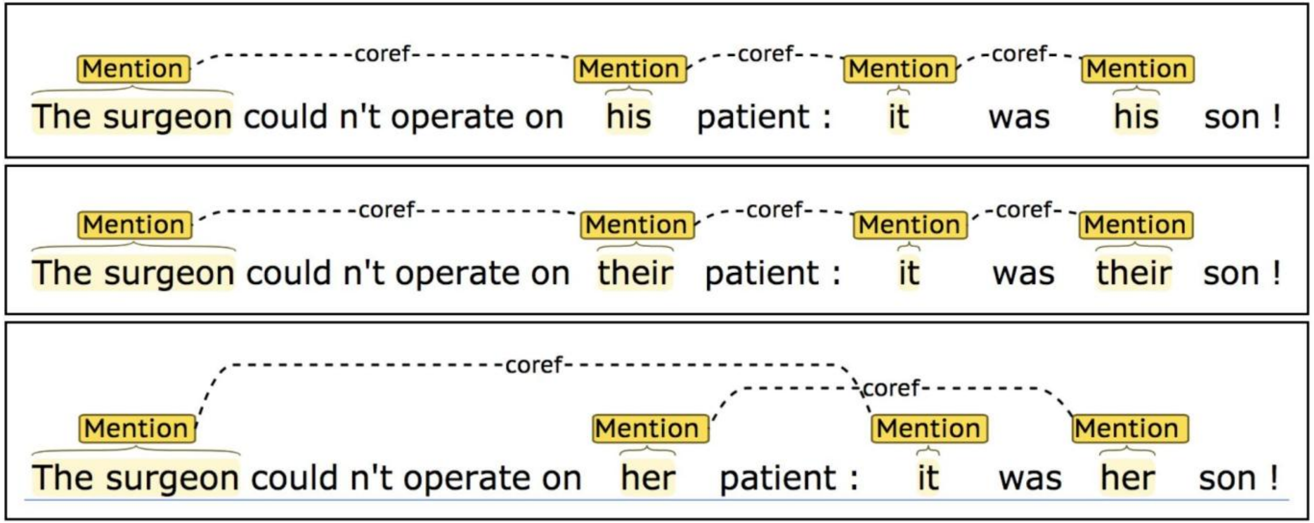
Rudinger et al, 2018

Gender Swapping in Coreference Resolution

Original sample

Gender swap

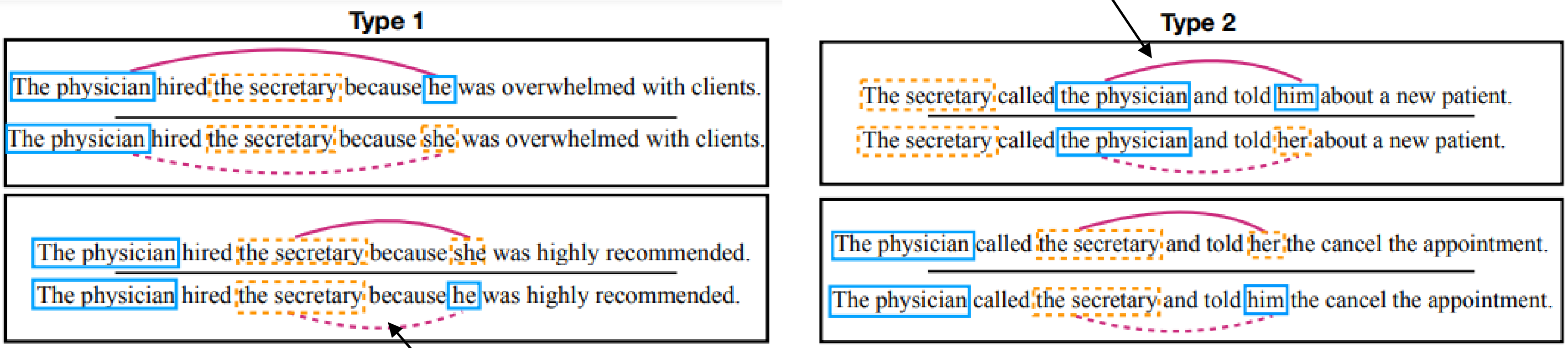
Gender swap



[Rudinger et al, 2018]

Gender Balanced Co-occurrence Test

Pro stereotypical



Anti stereotypical

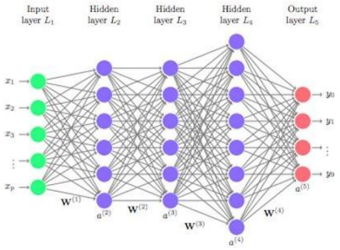
Type 1: [entity1] [interacts with] [entity2] [conjunction] [pronoun] [circumstances]

Type 2: [entity1] [interacts with] [entity2] and then [interacts with] [pronoun] for [circumstances].

Word Embeddings

- An Essential Part of Deep NLP Models
  - Classifications (e.g., Sentiment Analysis)
  - Text Generation (e.g., translation, summarization)
  - Text Retrieval (e.g., Question Answering)
  - Visual-Language Representations (e.g., Image Captioning)

Text → Look Ups → Word Embedding →

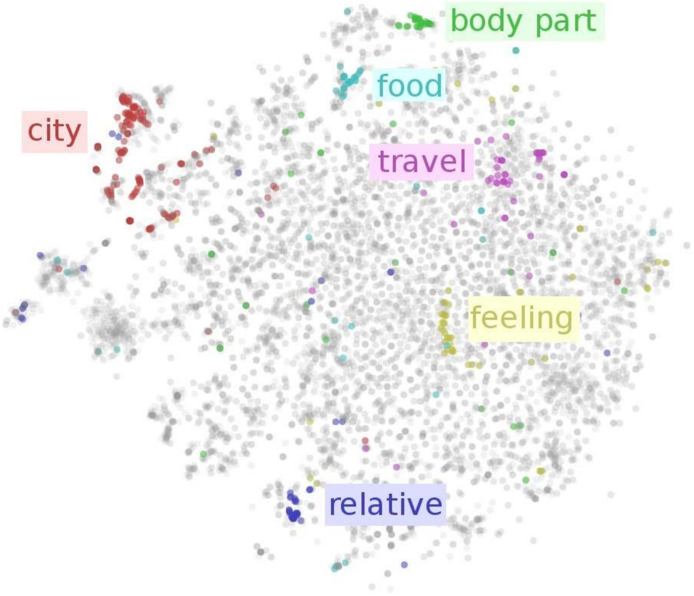


Discrete Space | Continuous Space | Neural Networks

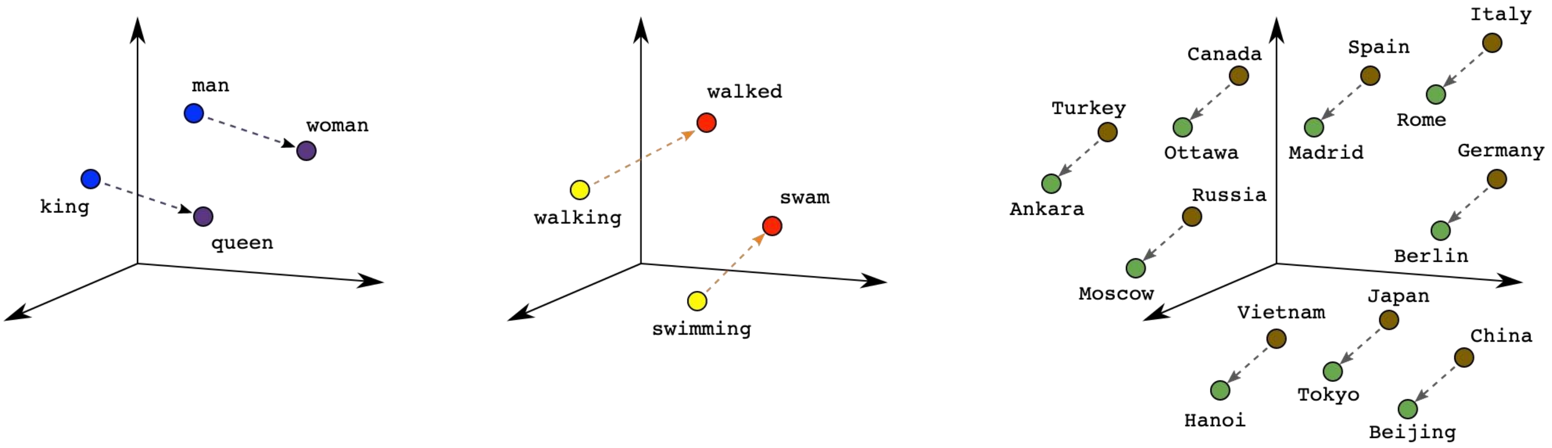
Word Embeddings

- Embedding Techniques
  - GloVe (Pennington et al, 2014)
  - Word2Vec (Rong et al, 2014)

- Trained Through A Proxy Task
  - Word proximity (GloVe)
  - SkipGram (Word2Vec)



Geometric Properties of Word Embeddings



Male-Female Verb Tense Country-Capital

Types of Gender Associations

- Definitional Gender Associations

$$\overrightarrow{\text{man}} - \overrightarrow{\text{woman}} \approx \overrightarrow{\text{king}} - \overrightarrow{\text{queen}}$$

- Stereotypical Gender Associations

$$\overrightarrow{\text{man}} - \overrightarrow{\text{woman}} \approx \overrightarrow{\text{computer programmer}} - \overrightarrow{\text{homemaker}}$$

Bolukbasi et al, 2016

Definitional and Stereotypical Associations

Gender stereotype *she*-*he* analogies.

sewing-carpentry	register-nurse-physician	housewife-shopkeeper
nurse-surgeon	interior designer-architect	softball-baseball
blond-burly	feminism-conservatism	cosmetics-pharmaceuticals
giggle-chuckle	vocalist-guitarist	petite-lanky
sassy-snappy	diva-superstar	charming-affable
volleyball-football	cupcakes-pizzas	hairdresser-barber

Gender appropriate *she*-*he* analogies.

queen-king	sister-brother	mother-father
waitress-waiter	ovarian cancer-prostate cancer	convent-monastery

Bolukbasi et al, 2016

Gender Subspace

$$\overrightarrow{\text{grandmother}} - \overrightarrow{\text{grandfather}} = \overrightarrow{\text{gal}} - \overrightarrow{\text{guy}} = g$$

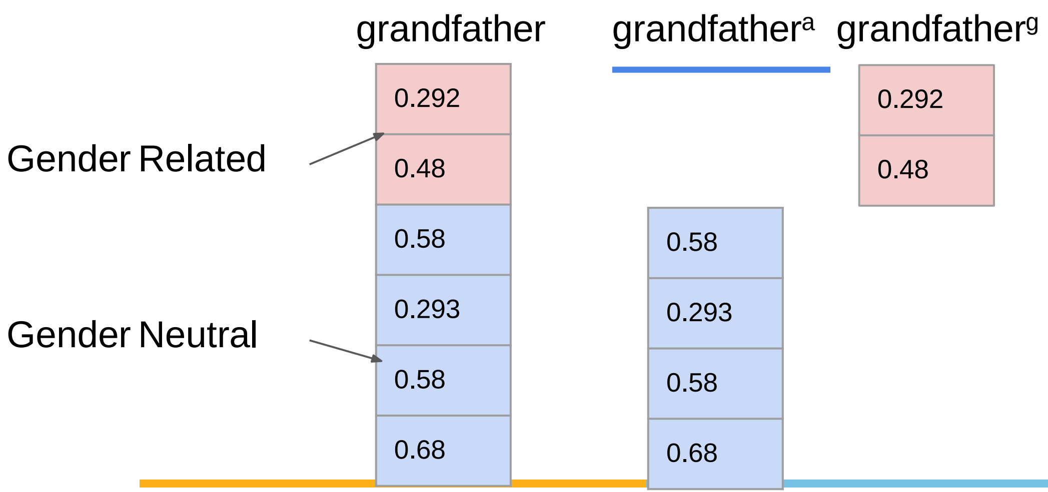
$$\begin{array}{rl} \overrightarrow{\text{she}} - \overrightarrow{\text{he}} & \qquad \overrightarrow{\text{daughter}} - \overrightarrow{\text{son}} \\ \overrightarrow{\text{her}} - \overrightarrow{\text{his}} & \qquad \overrightarrow{\text{mother}} - \overrightarrow{\text{father}} \\ \overrightarrow{\text{woman}} - \overrightarrow{\text{man}} & \qquad \overrightarrow{\text{gal}} - \overrightarrow{\text{guy}} \\ \overrightarrow{\text{Mary}} - \overrightarrow{\text{John}} & \qquad \overrightarrow{\text{girl}} - \overrightarrow{\text{boy}} \\ \overrightarrow{\text{herself}} - \overrightarrow{\text{himself}} & \qquad \overrightarrow{\text{female}} - \overrightarrow{\text{male}} \end{array}$$

Bolukbasi et al, 2016

Gender-Neutral Word Embeddings

- Decompose Word Embeddings Into Gender-Related and Gender-Neural Parts

$$w = [w^{(a)}; w^{(g)}]$$



Zhao et al, 2018

Gender-Neutral Word Embeddings

- Fine-tuning Word Embeddings Using Debiasing Regularizers

$$J = \underbrace{J_G}_{\text{Glove Loss Function}} + \underbrace{\lambda_d J_D}_{\text{Regulate Gender-related Words}} + \underbrace{\lambda_e J_E}_{\text{Regulate All Other Words}}$$

Glove
Loss Function

Regulate
Gender-related
Words

Regulate All Other
Words

Ω_F
Female Seed Words

Ω_N
All Other Words

Ω_M
Male Seed Words

[Zhao et al, 2018]

Gender-Neutral Word Embeddings

- Fine-tuning Word Embeddings Using Debiasing Regularizers

$$J = J_G + \lambda_d \underline{J_D} + \lambda_e J_E$$

Regulate
Gender-related
Words

Ω_F
Female Seed Word

Push Toward Extremes
On Gender Dimensions

Ω_M
Male Seed Word

$w^{(g)}$ - Gender-related Components
$w^{(a)}$ - Gender-neutral Components

$$J_D = -\left\| \sum_{w\in\Omega_M} w^{(g)} - \sum_{w\in\Omega_F} w^{(g)} \right\|_1 + \sum_{w\in\Omega_M} \left\| 1 - w^{(g)} \right\|_2^2 + \sum_{w\in\Omega_F} \left\| -1 - w^{(g)} \right\|_2^2$$

Zhao et al, 2018

Gender-Neutral Word Embeddings

- Fine-tuning Word Embeddings Using Debiasing Regularizers

$$J = J_G + \lambda_d J_D + \underline{\lambda_e J_E}$$

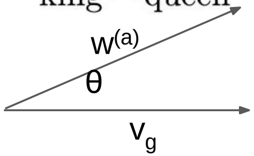
Regulate All Other Words

$$\overrightarrow{\text{man}} - \overrightarrow{\text{woman}}$$

$$\overrightarrow{\text{king}} - \overrightarrow{\text{queen}}$$

$$v_g = \frac{1}{|\Omega'|} \sum_{(w_m, w_f) \in \Omega_M, \Omega_F} (w_m^{(a)} - w_f^{(a)})$$

Gender Subspace

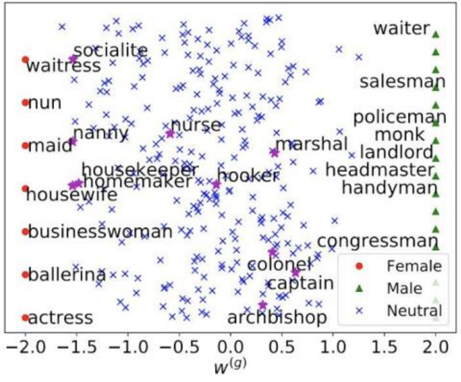


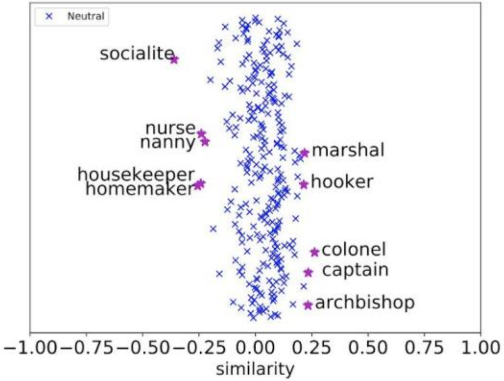
$$J_E = \sum_{w \in \Omega_N} \left(v_g^T w^{(a)} \right)^2$$

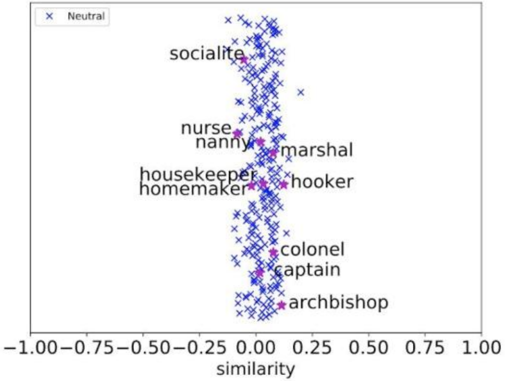
$w^{(g)}$ - Gender-related Components
$w^{(a)}$ - Gender-neutral Components

Zhao et al, 2018

Gender Attribute Separation







$w^{(g)}$ of All Occupations

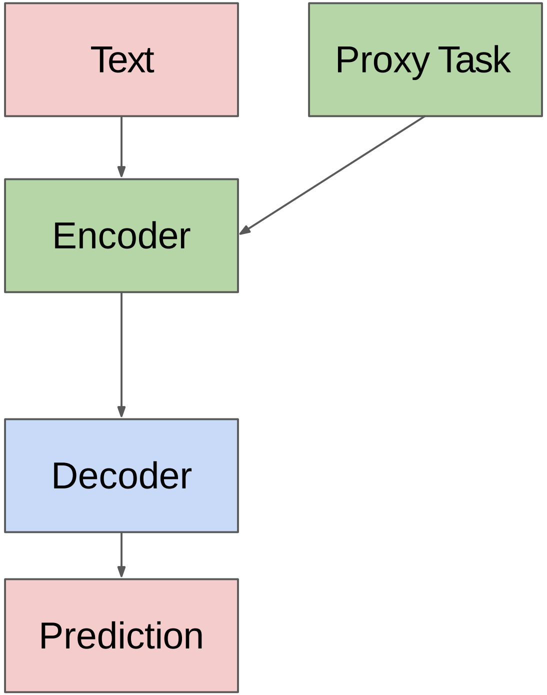
$w^{(a)}$ of GloVe for Gender Neutral Occupations

$w^{(a)}$ of Gender-Neutral GloVE for Gender Neutral Occupations

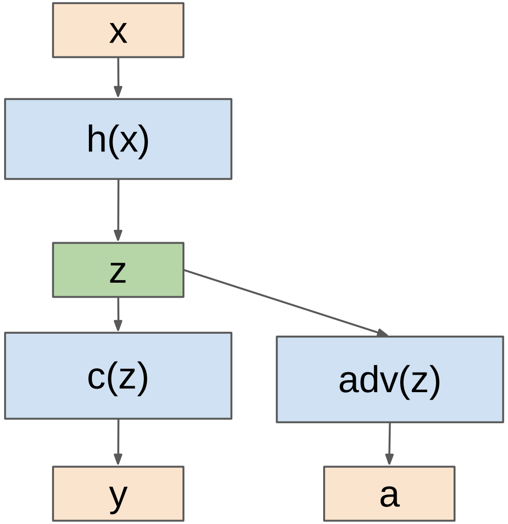
$w^{(g)}$ - Gender-related Components
$w^{(a)}$ - Gender-neutral Components

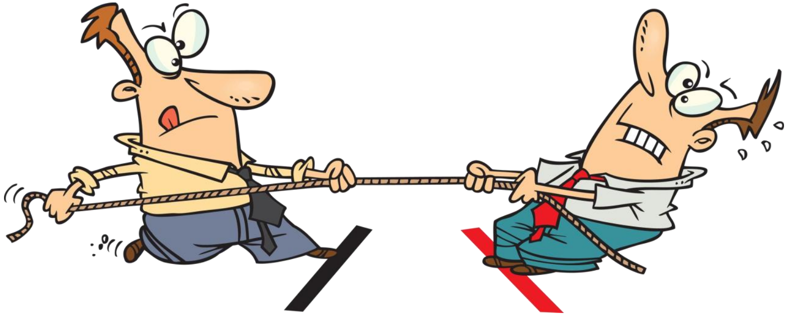
The Use of Pre-trained NLP Encoders

- Pre-trained Encoders Are Widely Used in NLP
  - Transfer information from a related domain
  - Boost performance on a small data set
  - Trained through a proxy task
- Pre-trained NLP Encoders
  - ELMO (Peters et al 2018)
  - BERT (Devlin et al, 2018)
  - XLNet (Yang et al, 2019)
- Can Pre-trained Encoders Be Biased?



Adversarial Learning



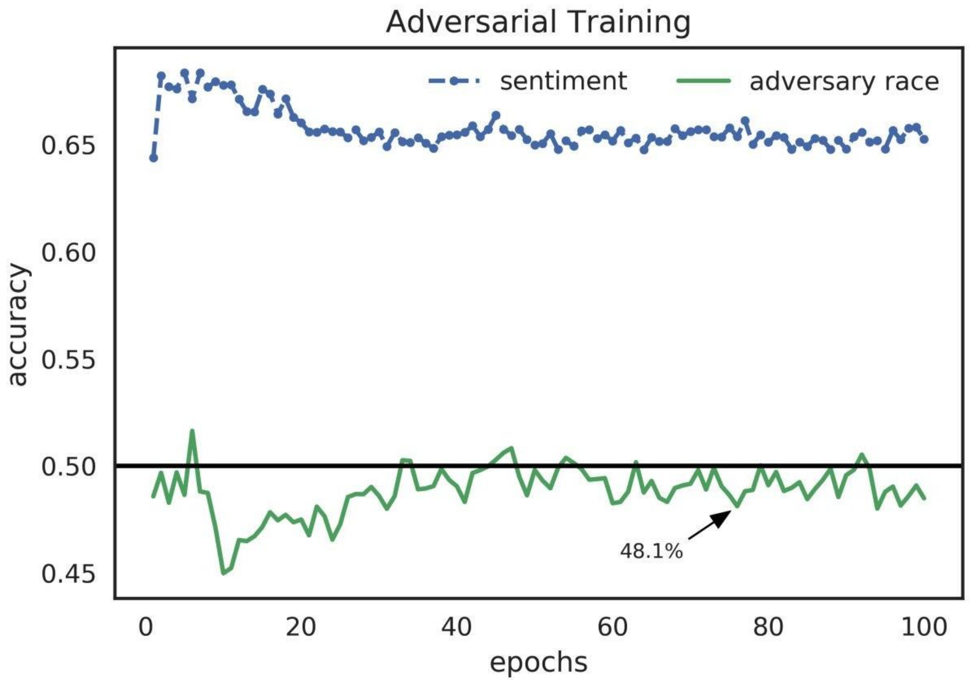


Predicting y

Reconstruct a

[Elazar et al, 2018](#)

Main Task and Adversary Accuracies



Elazar et al, 2018

Beefing Up the Adversary

- Increase the Capacity of the Adversary
  - Model Capacity
  - Weight on Loss
  - Ensemble

		DIAL			PAN16					
Method	Parameter	Sentiment	Race	Δ	Mention	Gender	Δ	Mention	Age	Δ
No Adversary Baseline	-	67.4	14.5	-	77.5	10.1	-	74.7	9.4	-
Standard Adversary	(300/1.0/1)	64.7	6.0	5.0	75.6	8.5	8.0	72.5	7.3	6.9
Adv-Capacity	500	64.1	6.7	5.2	73.8	8.1	6.7	71.4	**4.3**	4.1
	1000	63.4	7.1	4.9	75.2	8.9	7.0	71.6	6.3	4.0
	2000	65.2	8.1	6.9	76.1	6.7	6.4	71.9	6.0	5.7
	5000	63.9	**6.2**	3.7	74.5	5.6	1.6	73.0	10.2	9.6
	8000	65.0	7.1	4.8	75.7	**5.4**	4.2	71.9	9.8	7.3
λ	0.5	63.9	**6.8**	6.2	75.6	7.8	6.8	73.1	4.8	3.4
	1.5	64.9	7.4	5.4	75.6	**4.9**	2.4	72.5	6.8	5.8
	2.0	64.2	7.3	5.9	76.0	-7.2	6.7	72.1	8.5	7.7
	3.0	65.8	10.2	10.1	73.7	6.4	6.1	72.5	-6.3	5.2
	5.0	50.0	-	-	73.6	6.5	5.7	69.0	**3.2**	2.9
Ensemble	2	62.4	7.4	5.4	74.8	6.4	5.0	72.8	8.8	8.3
	3	66.5	6.5	5.0	75.3	4.9	3.1	72.1	6.7	6.0
	5	63.8	**4.8**	2.6	74.3	**4.1**	3.0	70.1	**5.7**	5.4

Δ - the difference between the attacker score and the corresponding adversary's accuracy

Elazar et al, 2018

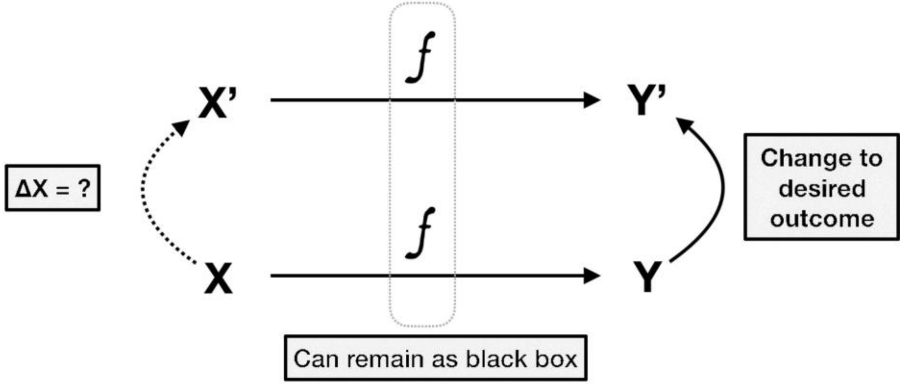
Fair Visual Representation

- Counterfactual Fairness
- Counterfactual Face Attribution
- Gender Equalized Image Captioning
- Adversarial Removal of Gender Features

Counterfactual Explanation

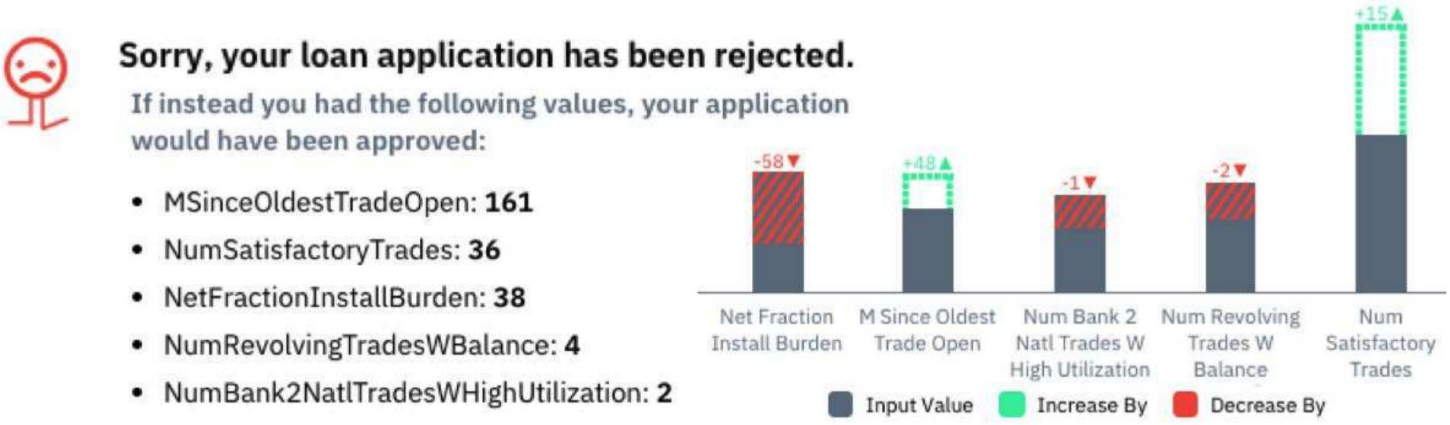
$$\underbrace{x'}_{\text{counterfactual example}} = arg \min_{x'} \lambda(\hat{f}(x') - \underbrace{y'}_{\text{desired outcome}})^2 + \underbrace{d(x, x')}_{\text{distance function}}$$

Increase λ while $|\hat{f}(x') - y'| > \varepsilon$



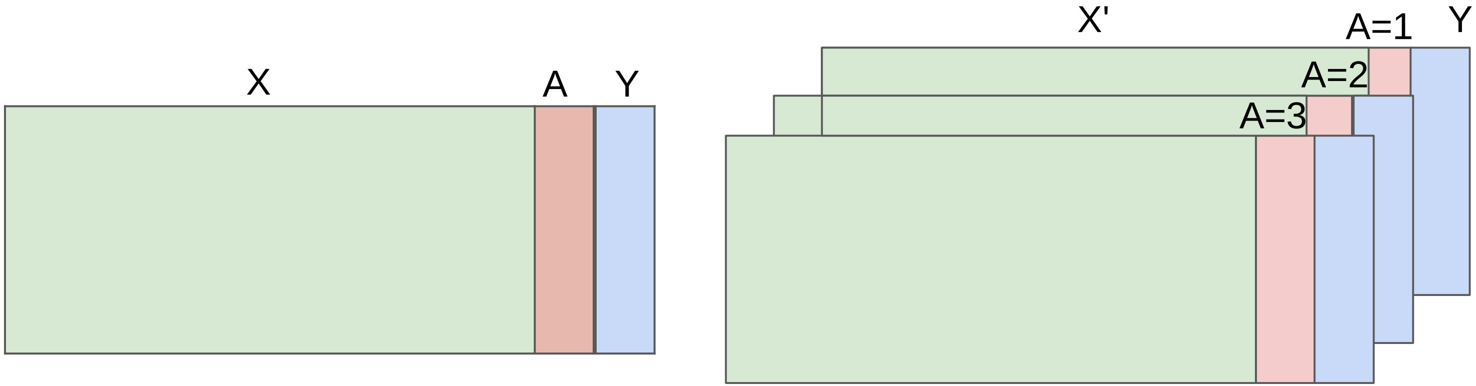
[Grath et al, 2018](https://)

Counterfactual Explanations



[Grath et al, 2018]

Counterfactual Fairness



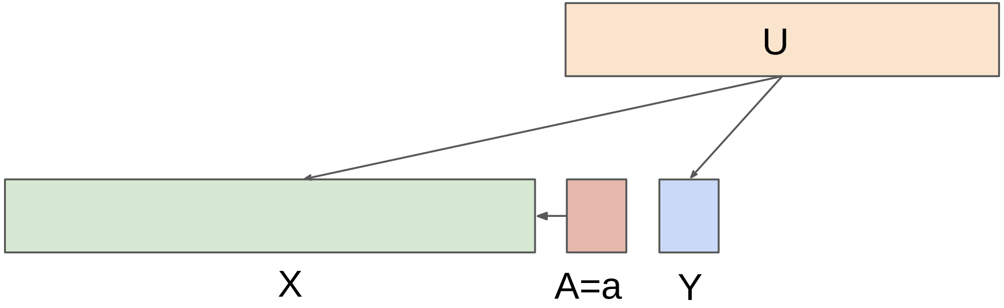
Real Examples

Counterfactual Examples

$$\underbrace{P(\hat{Y}_{A\leftarrow a}(U)=y \mid X=x, A=a)}_{\text{Real Examples}} = \underbrace{P(\hat{Y}_{A\leftarrow a'}(U)=y \mid X=x, A=a)}_{\text{Counterfactual Examples}}$$

Kusner et al, 2017

Causal View of Counterfactual Fairness

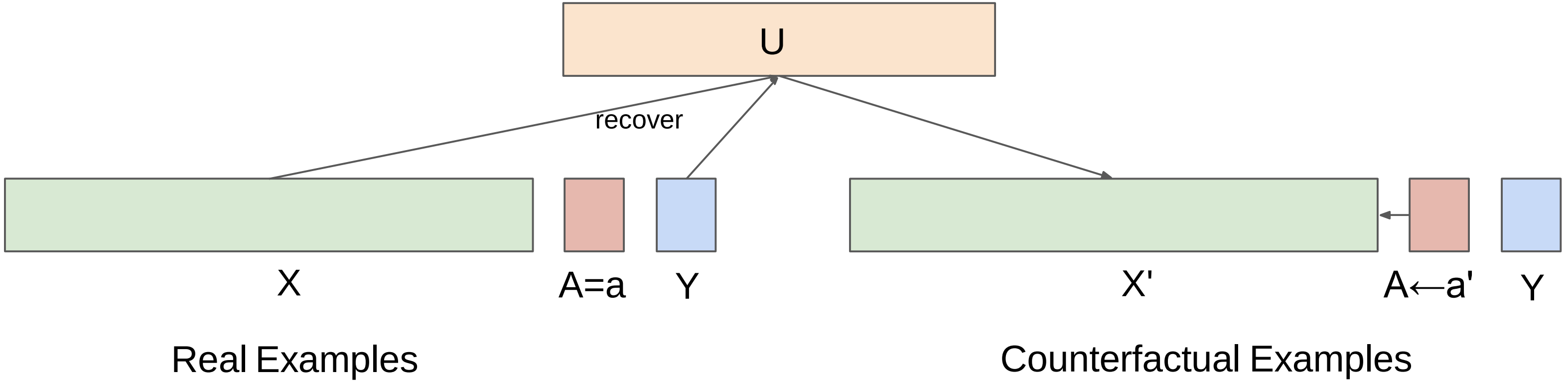


Real Examples

$$\underbrace{P(\hat{Y}_{A \leftarrow a}(U) = y \mid X = x, A = a)}_{\text{Real Examples}} = \underbrace{P(\hat{Y}_{A \leftarrow a'}(U) = y \mid X = x, A = a)}_{\text{Counterfactual Examples}}$$

[Kusner et al, 2017]

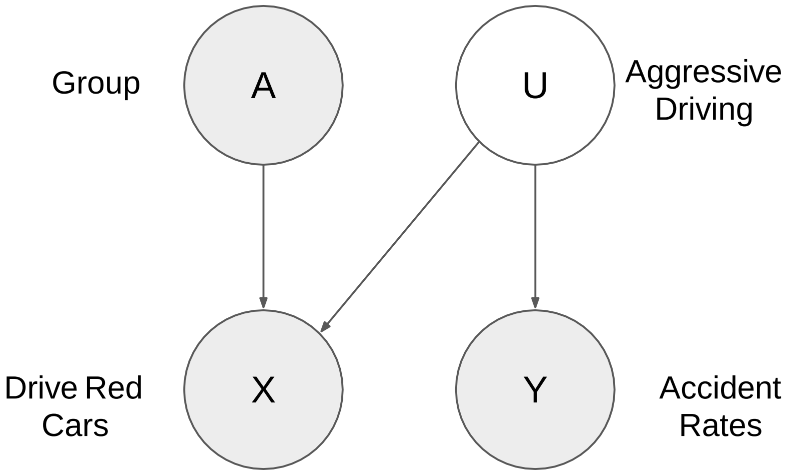
Causal View of Counterfactual Fairness

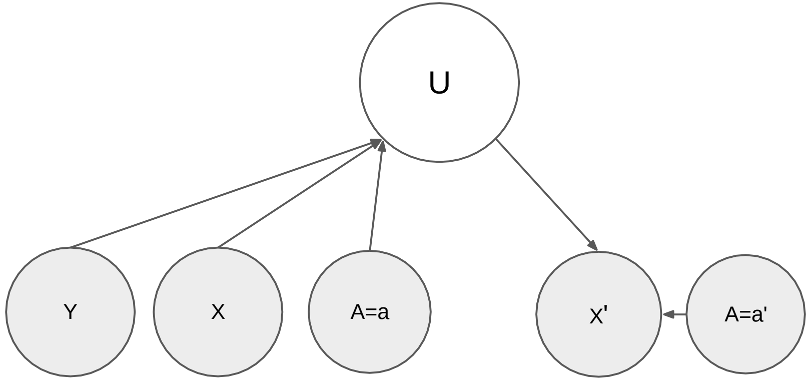


$$\underbrace{P(\hat{Y}_{A\leftarrow a}(U) = y \mid X = x, A = a)}_{\text{Real Examples}} = \underbrace{P(\hat{Y}_{A\leftarrow a'}(U) = y \mid X = x, A = a)}_{\text{Counterfactual Examples}}$$

Kusner et al, 2017

Causal Perspective of the Red Car Problem





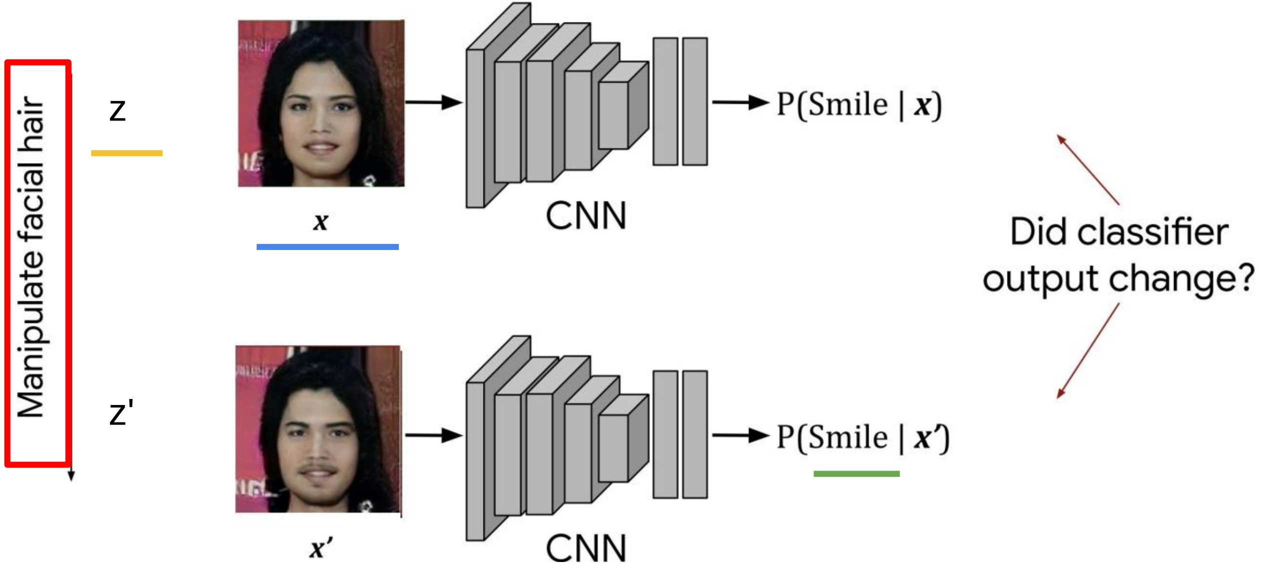
- Observed Variables
- Latent (Unobserved) Variables

$$P(\hat{Y}_{A\leftarrow a}\,(U) = y \mid X = x, A = a) = P(\hat{Y}_{A\leftarrow a'}(U) = y \mid X = x, A = a)$$

Kusner et al, 2017

Counterfactual Face Attribution

- Evaluate the Counterfactual Fairness of Face Recognition Systems

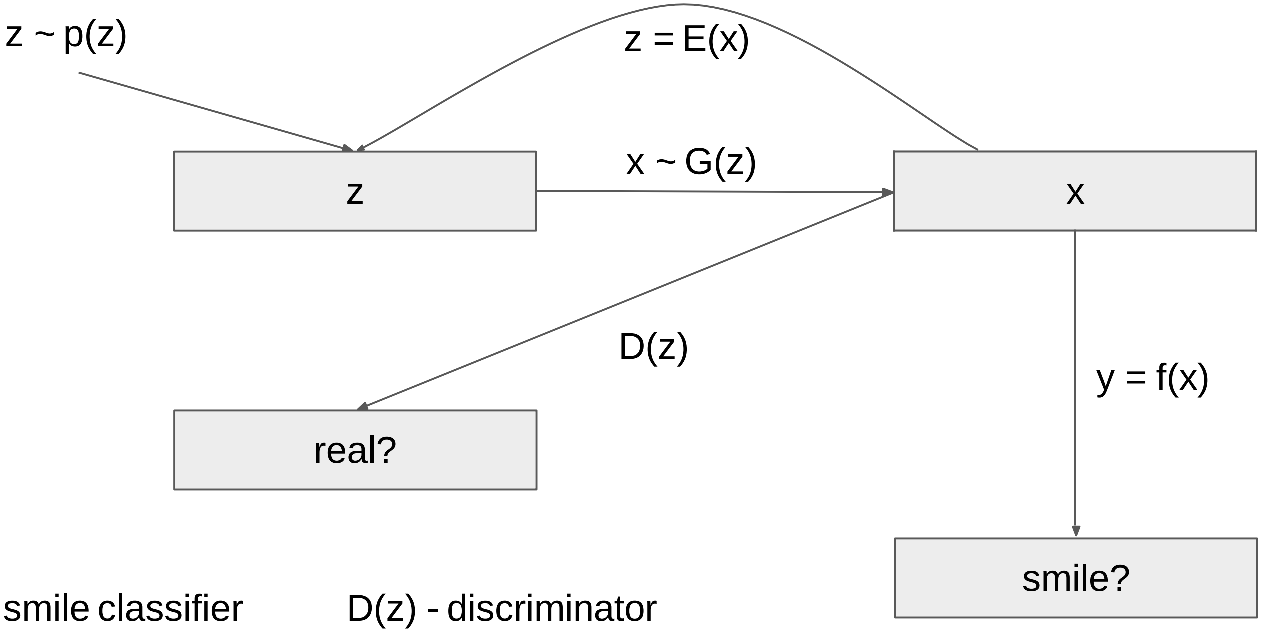


$$P(\hat{Y}_{A\leftarrow a}(U) = y \mid X = x, A = a) = P(\hat{Y}_{A\leftarrow a'}(U) = y \mid X = x, A = a)$$

CelebA Dataset

[Liu et al, 2015]

Model Architecture



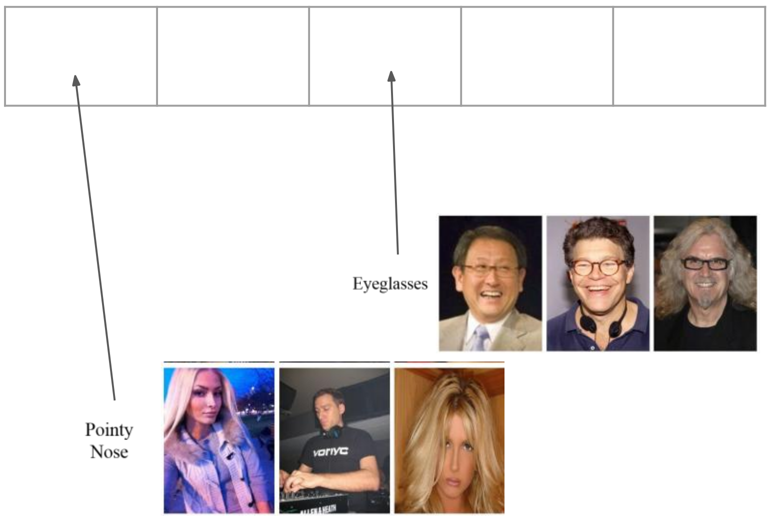
f(x) - smile classifier
G(z) - generator

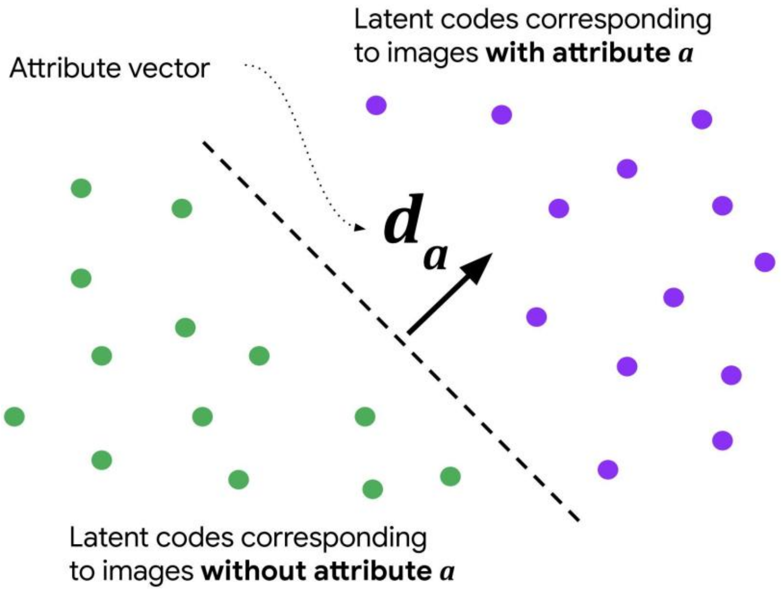
D(z) - discriminator
E (x) - encoder

[Denton et al, 2019]()

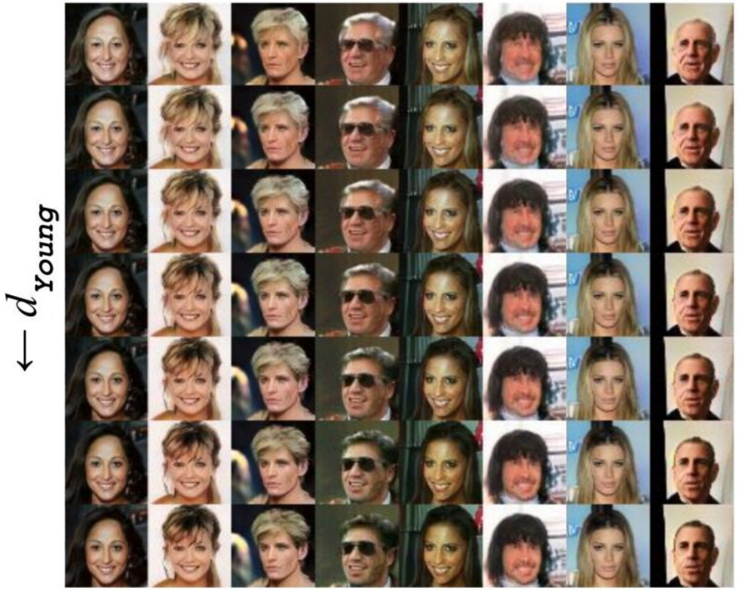
Latent Code Attribution

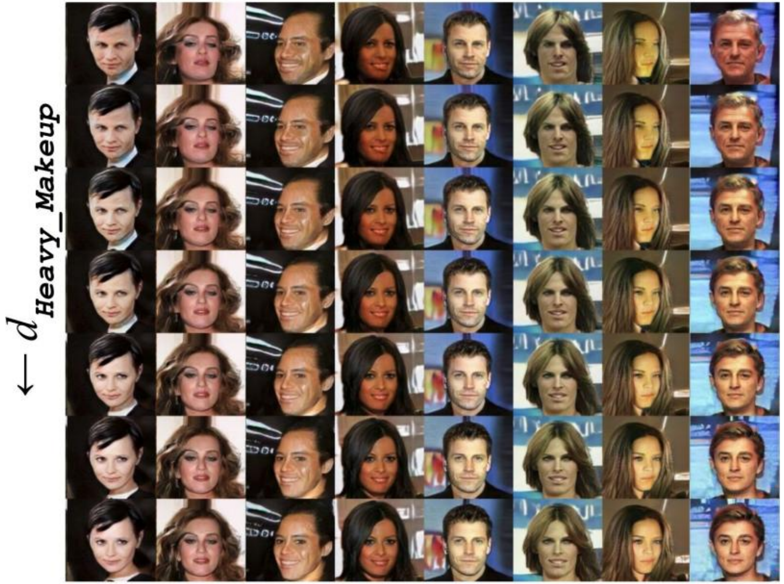
latent code z





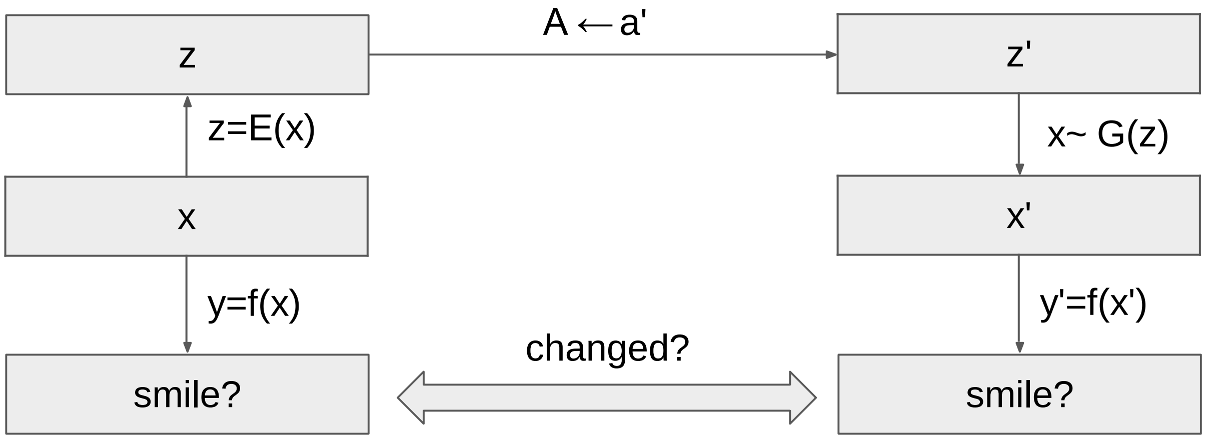
Latent Code Manipulation





[Denton et al, 2019]

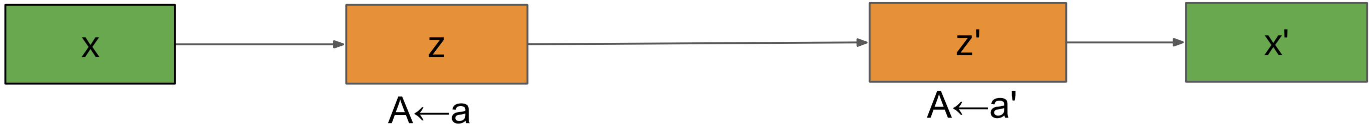
Counterfactual Fairness Assessment



$$P(\hat{Y}_{A\leftarrow a}(U) = y \mid X = x, A = a) = P(\hat{Y}_{A\leftarrow a'}(U) = y \mid X = x, A = a)$$

Denton et al, 2019

Sensitivity



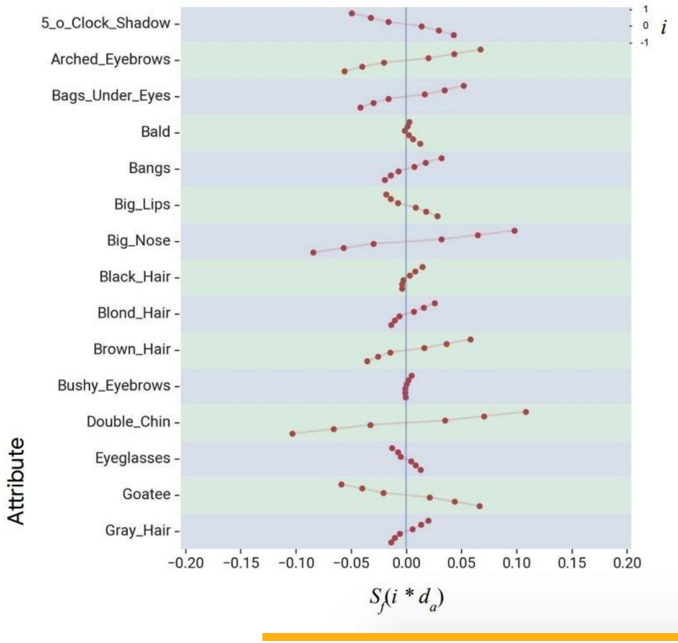
$$S_f(d) = \mathbb{E}_{z \sim p(z)}[f(G(z+d)) - f(G(z))]$$

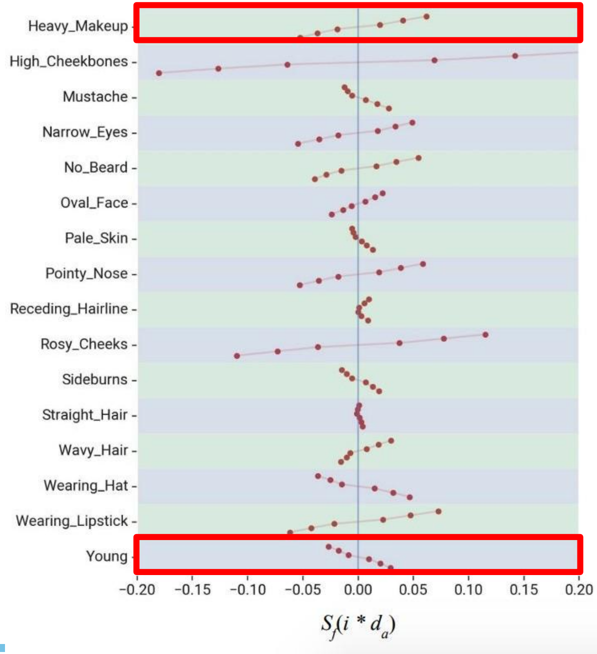
f(x) - smile classifier
G(z) - generator
D(z) - discriminator

$$P(\hat{Y}_{A\leftarrow a}(U) = y \mid X = x, A = a) = P(\hat{Y}_{A\leftarrow a'}(U) = y \mid X = x, A = a)$$

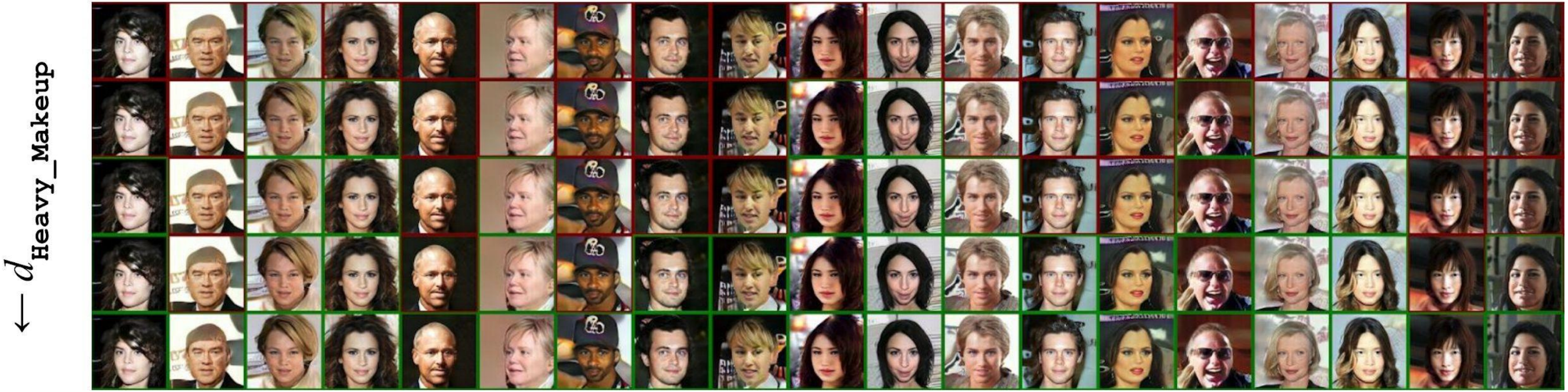
Sensitivity Results

$$S_f(d) = \mathbb{E}_{z \sim p(z)}[f(G(z+d)) - f(G(z))]$$

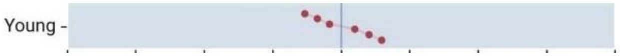




Heavy Makeup



Young



Directional Sensitivity

$$S_y^{0\to1}(d) = \mathbb{E}_{z\sim p(z)|y(G(z))=0}\, \mathbb{I}[y(G(z+d))! = y(G(z))]$$

from "not smiling" to "smiling"

$$S_y^{1\to0}(d) = \mathbb{E}_{z\sim p(z)|y(G(z))=1}\, \mathbb{I}[y(G(z+d))! = y(G(z))]$$

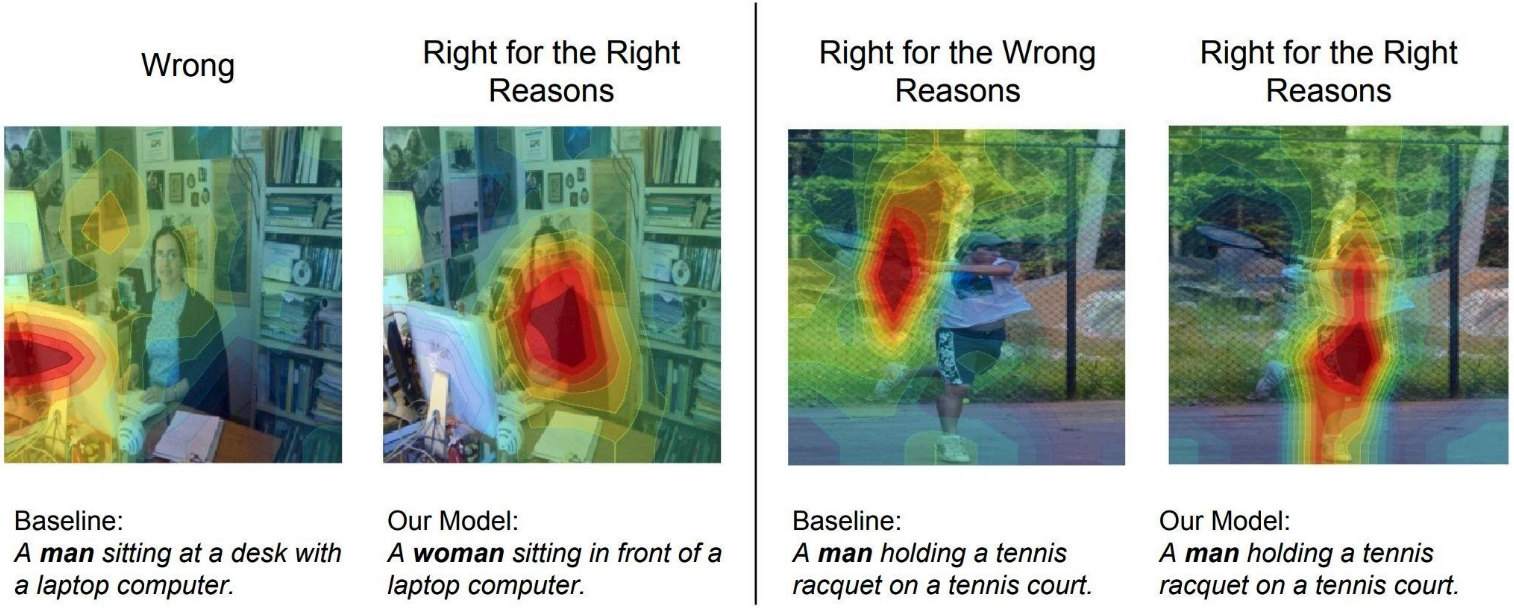
from "smiling" to "not smiling"

$$S_f(d) = \mathbb{E}_{z\sim p(z)}[f(G(z+d)) - f(G(z))]$$

Sensitivity Results

CelebA attribute defining d_a	$S_y^{1\to0}(d_a)$	$S_y^{0\to1}(d_a)$
Young	**7.0%**	2.6%
5_o_Clock_Shadow	**11.8%**	2.2%
Goatee	**12.4%**	0.9%
No_Beard	0.8%	**11.8%**
Heavy_Makeup	1.6%	**12.4%**
Wearing_Lipstick	1.7%	**16.3%**

The Equalizer Model

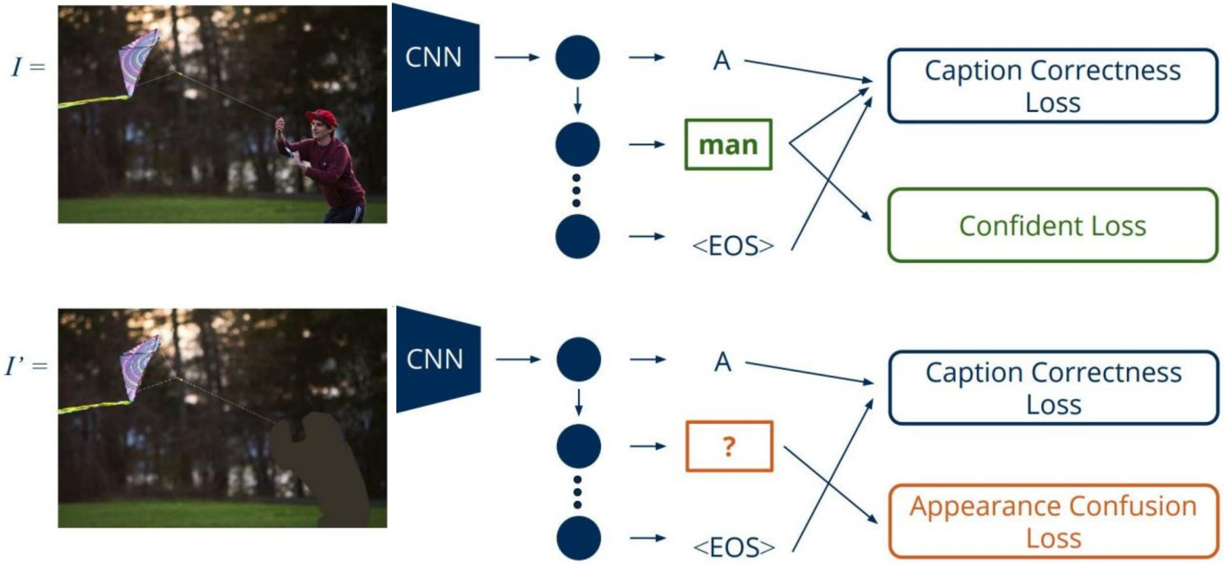


[Burns et al, 2019]

The Basic Idea

Original Image

Annotated Gender Neutral Image



[Burns et al, 2019]

The Equalizer Model

$$\mathcal{L} = \alpha\mathcal{L}^{CE} + \beta\mathcal{L}^{AC} + \mu\mathcal{L}^{Con}$$

Cross Entropy Loss

$$\mathcal{L}^{CE} = -\frac{1}{N}\sum_{n=0}^{N}\sum_{t=0}^{T}\log(p(w_t|w_{0:t-1}, I))$$

Appearance Confusing Loss on the gender neutral image

Confidence Loss on the original image

Burns et al, 2019

Appearance Confusing Objective

$$\mathcal{L}^{AC} = \frac{1}{N}\sum_{n=0}^{N}\sum_{t=0}^{T}\mathbb{1}(w_t \in \mathcal{G}_w \cup \mathcal{G}_m)\mathcal{C}(\tilde{w}_t, I')$$

$$\mathcal{C}(\tilde{w}_t, I') = |\sum_{g_w \in \mathcal{G}_w} p(\tilde{w}_t = g_w | w_{0:t-1}, I') - \sum_{g_m \in \mathcal{G}_m} p(\tilde{w}_t = g_m | w_{0:t-1}, I')|$$

Push Toward Extremes

$p(\tilde{w}_t = g_w | w_{0:t-1}, I')$ ⟺ $p(\tilde{w}_t = g_m | w_{0:t-1}, I')$

$I' =$

$\mathcal{G}_w$ - set of words for woman

$\mathcal{G}_m$ - set of words for man

[Burns et al, 2019]

Confidence Objective

$$\mathcal{L}^{Con} = \frac{1}{N}\sum_{n=0}^{N}\sum_{t=0}^{T}(\mathbb{1}(w_t \in \mathcal{G}_w)\mathcal{F}^W(\tilde{w}_t, I) + \mathbb{1}(w_t \in \mathcal{G}_m)\mathcal{F}^M(\tilde{w}_t, I))$$

$$\mathcal{F}^W(\tilde{w}_t, I) = \frac{\sum_{g_m \in \mathcal{G}_m} p(\tilde{w}_t = g_m | w_{0:t-1}, I)}{(\sum_{g_w \in \mathcal{G}_w} p(\tilde{w}_t = g_w | w_{0:t-1}, I)) + \epsilon}$$

$I =$ 

$\mathcal{G}_w$ - set of words for woman
$\mathcal{G}_m$ - set of words for man

Burns et al, 2019

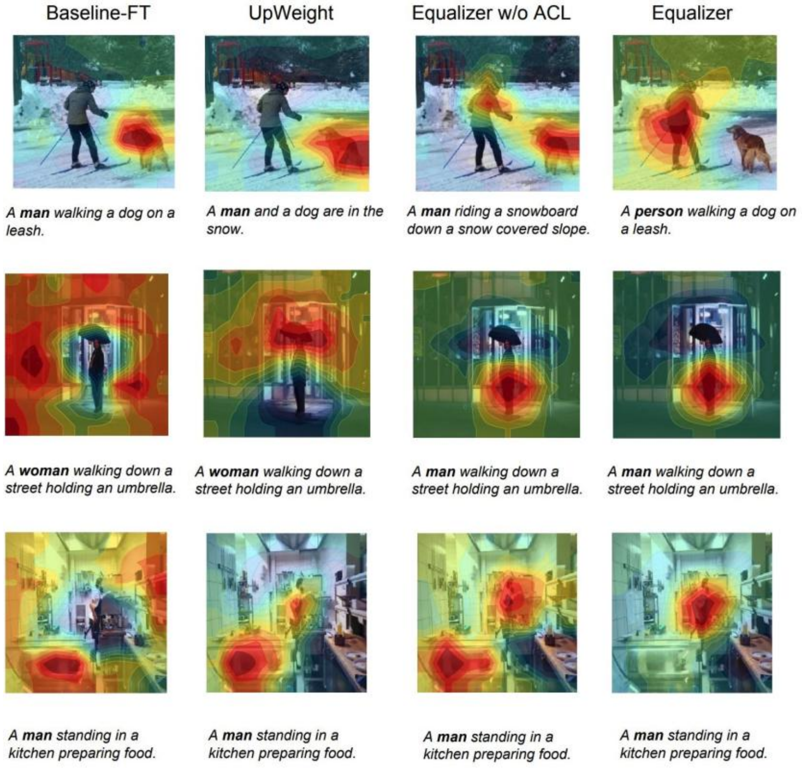
Results

Model	MSCOCO-Bias Error	MSCOCO-Bias Ratio Δ	MSCOCO-Balanced Error	MSCOCO-Balanced Ratio Δ
Baseline-FT	12.83	0.15	19.30	0.51
Balanced	12.85	0.14	18.30	0.47
UpWeight	13.56	0.08	16.30	0.35
Equalizer w/o ACL	7.57	0.04	10.10	0.26
Equalizer w/o Conf	9.62	0.09	13.90	0.40
Equalizer	**7.02**	**-0.03**	**8.10**	**0.13**

Baseline-FT - basic LSTM attention model (Xu et al, 2015)
Balanced - resampled dataset to have balanced gender ratio
UpWeight - reweighting
Δ- change to the gender ratio compared to the dataset

Burns et al, 2019

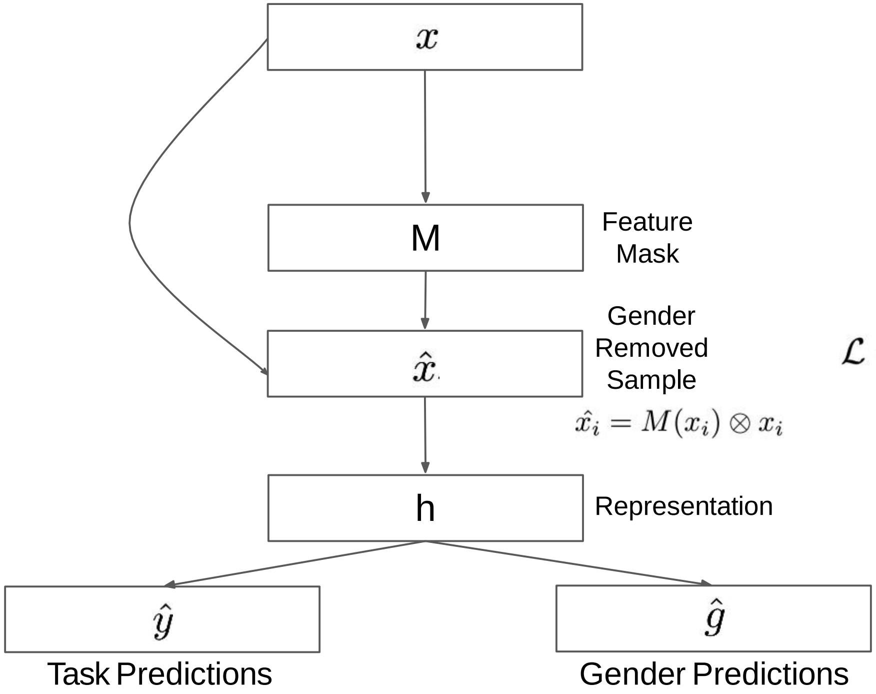
Results


Baseline-FT
UpWeight
Equalizer w/o ACL
Equalizer
A **man** walking a dog on a leash.
A **man** and a dog are in the snow.
A **man** riding a snowboard down a snow covered slope.
A **person** walking a dog on a leash.
A **woman** walking down a street holding an umbrella.
A **woman** walking down a street holding an umbrella.
A **man** walking down a street holding an umbrella.
A **man** walking down a street holding an umbrella.
A **man** standing in a kitchen preparing food.
A **man** standing in a kitchen preparing food.
A **man** standing in a kitchen preparing food.
A **man** standing in a kitchen preparing food.

Adversarial Removal of Gender Features

Wang et al, 2019

Model Architecture



$$\mathcal{L} = \sum_i \beta |x_i - \hat{x}_i| + \mathcal{L}_p(pred(h(\hat{x}_i)), y_i) - \lambda \mathcal{L}_c(c(h(\hat{x}_i)), g_i)$$

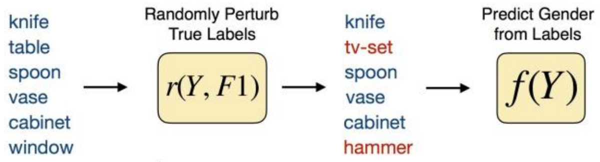
Reconstruction Loss   Task Loss   Adversarial Loss

[Wang et al, 2019]

Evaluate Sensitive Information Leakage

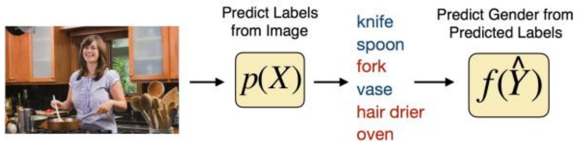
- Train an attacker f(y) that reverse engineer the gender information

DATASET LEAKAGE @ F1



$$\lambda_D(a) = \frac{1}{|\mathcal{D}|}\sum\nolimits_{(Y_i,g_i)\in\mathcal{D}} \mathbb{1}[f(r(Y_i,a)) == g_i]$$

MODEL LEAKAGE @ F1



$$\lambda_M(a) = \frac{1}{|\mathcal{D}|}\sum\nolimits_{(\hat{Y}_i,g_i)\in\mathcal{D}} \mathbb{1}\,[f(r(\hat{Y}_i,a) == g_i)]$$

Data Resampling

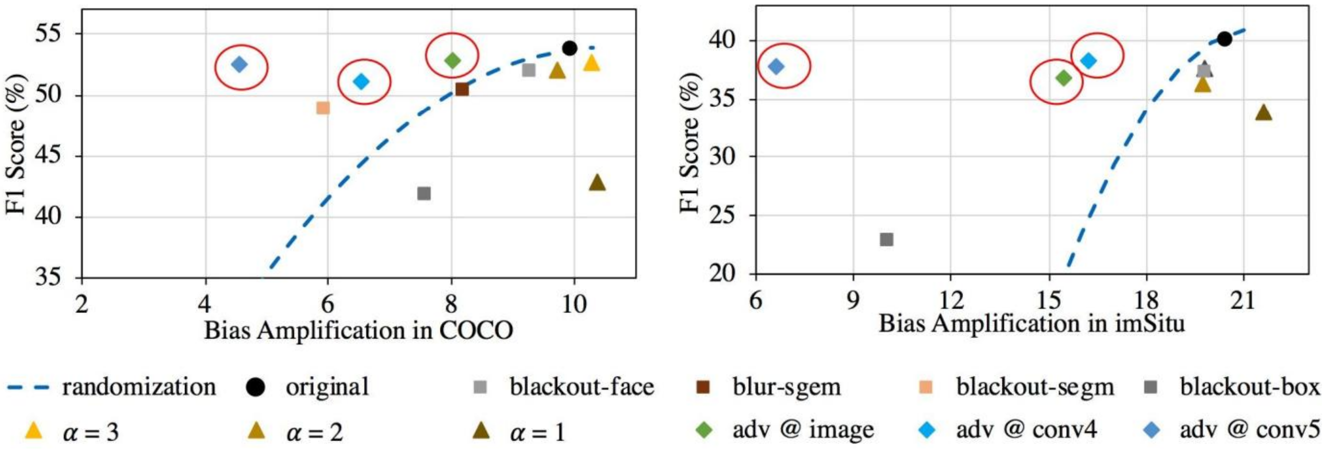
$$\forall y : 1/\alpha < \#(m,y)/\#(w,y) < \alpha$$

Bias Amplification

$$\Delta = \lambda_M(a) - \lambda_D(a)$$

Wang et al, 2019

Accuracy and Bias Results

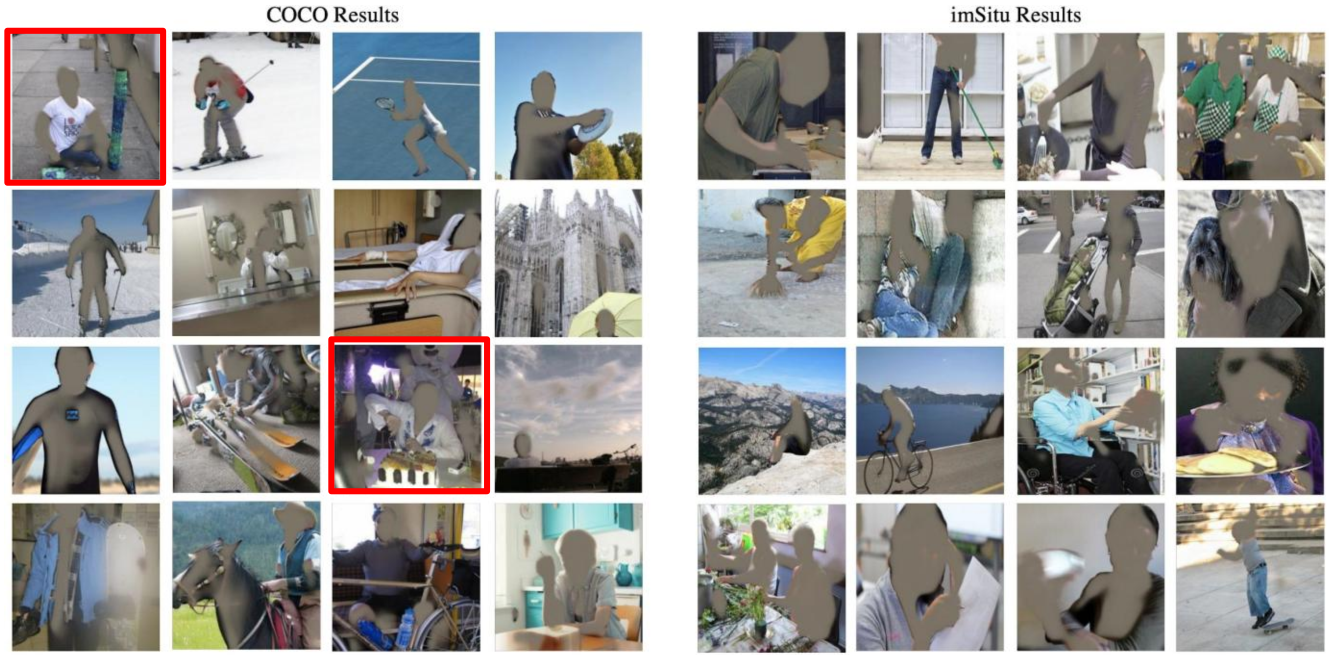


original - no debiasing mask
random - adding random noise

blackout-face - black out using a face detector
blur-sgem - black out using ground truth segmentation
blackout-box - blackout using bounding boxes

Wang et al, 2019

Qualitative Results



Wang et al, 2019